

$$15. \begin{vmatrix} b^2c^2 + a^2 & bc + a & 1 \\ c^2a^2 + b^2 & ca + b & 1 \\ a^2b^2 + c^2 & ab + c & 1 \end{vmatrix}$$

$$16. \begin{vmatrix} a^2 & b^2 & c^2 & d^2 \\ a & b & c & d \\ 1 & 1 & 1 & 1 \\ bcd & cda & dab & abc \end{vmatrix}$$

(Andhra, 1999)

$$17. \text{ If } a + b + c = 0, \text{ solve } \begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$$

$$18. \text{ Solve the equation } \begin{vmatrix} x+1 & 2x+1 & 3x+1 \\ 2x & 4x+3 & 6x+3 \\ 4x+1 & 6x+4 & 8x+4 \end{vmatrix} = 0.$$

$$19. \text{ Show that } \begin{vmatrix} b^2 + c^2 & ab & ac \\ ba & c^2 + a^2 & bc \\ ca & cb & a^2 + b^2 \end{vmatrix} = \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}^2 = 4a^2b^2c^2.$$

2.4 MATRICES

(1) **Definition.** A system of mn numbers arranged in a rectangular formation along m rows and n columns and bounded by the brackets $[]$ is called an m by n **matrix**; which is written as $m \times n$ matrix. A matrix is also denoted by a single capital letter.

Thus

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots a_{1j} & \dots a_{1n} \\ a_{21} & a_{22} & \dots a_{2j} & \dots a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots a_{ij} & \dots a_{in} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{mj} & \dots a_{mn} \end{bmatrix}$$

is a matrix of order mn . It has m rows and n columns. Each of the mn numbers is called an element of the matrix.

To locate any particular element of a matrix, the elements are denoted by a letter followed by two suffixes which respectively specify the rows and columns. Thus a_{ij} is the element in the i -th row and j -th column of A . In this notation, the matrix A is denoted by $[a_{ij}]$.

A matrix should be treated as a single entity with a number of components, rather than a collection of numbers. For example, the coordinates of a point in solid geometry, are given by a set of three numbers which can be represented by the matrix $[x, y, z]$. Unlike a determinant, a matrix cannot reduce to a single number and the question of finding the value of a matrix never arises. The difference between a determinant and a matrix is brought out by the fact that an interchange of rows and columns does not alter the determinant but gives an entirely different matrix.

(2) Special matrices

Row and column matrices. A matrix having a single row is called a row matrix, e.g., $[1 \ 3 \ 5 \ 7]$.

A matrix having a single column is called a column matrix, e.g., $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$

Row and column matrices are sometimes called *row vectors* and *column vectors*.

Square matrix. A matrix having n rows and n columns is called a square matrix of order n .

The determinant having the same elements as the square matrix A is called the *determinant of the matrix* and is denoted by the symbol $|A|$. For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}, \text{ then } |A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

The diagonal of this matrix containing the elements 1, 3, 5 is called the *leading or principal diagonal*. The sum of the diagonal elements of a square matrix A is called the **trace** of A .

A square matrix is said to be **singular** if its determinant is zero otherwise **non-singular**.

Diagonal matrix. A square matrix all of whose elements except those in the leading diagonal, are zero is called a diagonal matrix.

A diagonal matrix whose all the leading diagonal elements are equal is called a scalar matrix. For example,

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

are the diagonal and scalar matrices respectively.

Unit matrix. A diagonal matrix of order n which has unity for all its diagonal elements, is called a unit matrix or an identity matrix of order n and is denoted by I_n . For example, unit matrix of order 3 is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Null matrix. If all the elements of a matrix are zero, it is called a null or zero matrix and is denoted by '0'; e.g.,

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ is a null matrix.}$$

Symmetric and skew-symmetric matrices. A square matrix $A = [a_{ij}]$ is said to be symmetric when $a_{ij} = a_{ji}$ for all i and j .

If $a_{ij} = -a_{ji}$ for all i and j so that all the leading diagonal elements are zero, then the matrix is called a skew-symmetric matrix. Examples of symmetric and skew-symmetric matrices are

$$\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & h & -g \\ -h & 0 & f \\ g & -f & 0 \end{bmatrix} \text{ respectively.}$$

Triangular matrix. A square matrix all of whose elements below the leading diagonal are zero, is called an upper triangular matrix. A square matrix all of whose elements above the leading diagonal are zero, is called a lower triangular matrix. Thus

$$\begin{bmatrix} a & h & g \\ 0 & b & f \\ 0 & 0 & c \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & -5 & 4 \end{bmatrix}$$

are upper and lower triangular matrices respectively.

2.5 MATRICES OPERATIONS

(1) Equality of Matrices

Two matrices A and B are said to equal if and only if

(i) they are of the same order

and (ii) each element of A is equal to the corresponding element of B .

(2) Addition and subtraction of matrices. If A, B be two matrices of the same order, then their sum $A + B$ is defined as the matrix each element of which is the sum of the corresponding elements of A and B .

$$\text{Thus,} \quad \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} + \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \\ c_3 & d_3 \end{bmatrix} = \begin{bmatrix} a_1 + c_1 & b_1 + d_1 \\ a_2 + c_2 & b_2 + d_2 \\ a_3 + c_3 & b_3 + d_3 \end{bmatrix}$$

Similarly, $A - B$ is defined as a matrix whose elements are obtained by subtracting the elements of B from the corresponding elements of A .

$$\text{Thus,} \quad \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} - \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 - c_1 & b_1 - d_1 \\ a_2 - c_2 & b_2 - d_2 \end{bmatrix}$$

Obs. 1. Only matrices of the same order can be added or subtracted.

2. Addition of matrices is commutative,

$$\text{i.e.,} \quad A + B = B + A.$$

3. Addition and subtraction of matrices is associative.

$$\text{i.e. } (A + B) - C = A + (B - C) = B + (A - C).$$

(3) Multiplication of matrix by a scalar. The product of a matrix A by a scalar k is a matrix whose each element is k times the corresponding elements of A .

$$\text{Thus, } k \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} = \begin{bmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \end{bmatrix}$$

The distributive law holds for such products, i.e., $k(A + B) = kA + kB$.

Obs. All the laws of ordinary algebra hold for the addition or subtraction of matrices and their multiplication by scalars.

Example 2.12. Find x, y, z and w given that

$$3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 5 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 6 & x+y \\ z+w & 5 \end{bmatrix}$$

$$\text{Solution. We have } \begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix} = \begin{bmatrix} x+6 & 5+x+y \\ -1+z+w & 2w+5 \end{bmatrix}$$

Equating the corresponding elements, we get

$$3x = x + 6, 3y = 5 + x + y, 3z = -1 + z + w, 3w = 2w + 5.$$

$$\text{or } 2x = 6, 2y = 5 + x, 2z = w - 1, w = 5$$

$$\text{Hence } x = 3, y = 4, z = 2, w = 5.$$

Example 2.13. Express $\begin{bmatrix} 3 & 5 & -7 \\ -8 & 11 & 4 \\ 13 & -14 & 6 \end{bmatrix}$ as the sum of a lower triangular matrix with zero leading diagonal and an upper triangular matrix.

$$\text{Solution. Let } L = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{bmatrix} \text{ be the lower triangular matrix with zero leading diagonal.}$$

$$\text{and } U = \begin{bmatrix} l & m & n \\ 0 & p & q \\ 0 & 0 & r \end{bmatrix} \text{ be the upper triangular matrix.}$$

$$\text{Then } \begin{bmatrix} 3 & 5 & -7 \\ -8 & 11 & 4 \\ 13 & -14 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{bmatrix} + \begin{bmatrix} l & m & n \\ 0 & p & q \\ 0 & 0 & r \end{bmatrix}$$

Equating corresponding elements from both sides, we obtain $3 = l, 5 = m, -7 = n, -8 = a, 11 = p, 4 = q, 13 = b, -14 = c, 6 = r$.

$$\text{Hence } L = \begin{bmatrix} 0 & 0 & 0 \\ -8 & 0 & 0 \\ 13 & -14 & 0 \end{bmatrix} \text{ and } U = \begin{bmatrix} 3 & 5 & -7 \\ 0 & 11 & 4 \\ 0 & 0 & 6 \end{bmatrix}$$

(4) Multiplication of matrices. Two matrices can be multiplied only when the number of columns in the first matrix is equal to the number of rows in the second. Such matrices are said to be **conformable**.

$$\text{For instance, the product } \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{bmatrix} \times \begin{bmatrix} l_1 & l_2 \\ m_1 & m_2 \\ n_1 & n_2 \end{bmatrix}$$

$$\text{is defined as the matrix } \begin{bmatrix} a_1l_1 + b_1m_1 + c_1n_1 & a_1l_2 + b_1m_2 + c_1n_2 \\ a_2l_1 + b_2m_1 + c_2n_1 & a_2l_2 + b_2m_2 + c_2n_2 \\ a_3l_1 + b_3m_1 + c_3n_1 & a_3l_2 + b_3m_2 + c_3n_2 \\ a_4l_1 + b_4m_1 + c_4n_1 & a_4l_2 + b_4m_2 + c_4n_2 \end{bmatrix}$$

In general, if $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & b_{2p} \\ \dots & \dots & \dots & \dots \\ b_{n1} & b_{n2} & \dots & b_{np} \end{bmatrix}$

two $m \times n$ and $n \times p$ conformable matrices, then their product is defined as the $m \times p$ matrix

$$AB = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1p} \\ c_{21} & c_{22} & \dots & c_{2p} \\ \dots & \dots & \dots & \dots \\ c_{m1} & c_{m2} & \dots & c_{mp} \end{bmatrix}$$

where $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \dots + a_{in}b_{nj}$, i.e., the element in the i th row and the j th column of the matrix AB obtained by weaving the i th row of A with j th column of B . The expression for c_{ij} is known as the *inner product* of the i th row with the j th column.

Post-multiplication and Pre-multiplication. In the product AB , the matrix A is said to be *post-multiplied* by the matrix B . Whereas in BA , the matrix A is said to be *pre-multiplied* by B . In one case the product may exist and in the other case it may not. Also the product in both cases may exist yet may or may not be equal.

Obs. 1. Multiplication of matrices is associative. i.e., $(AB)C = A(BC)$ provided A, B are conformable for the product AB and B, C are conformable for the product BC . (Ex. 2.16).

Obs. 2. Multiplication of matrices is distributive. i.e., $A(B + C) = AB + AC$ provided A, B are conformable for the product AB and A, C are conformable for the product AC .

Obs. 3. Power of a matrix. If A be a square matrix, then the product AA is defined as A^2 . Similarly, we define higher powers of A . i.e., $A \cdot A^2 = A^3, A^2 \cdot A^2 = A^4$, etc.

If $A^2 = A$, then the matrix A is called *idempotent*.

Example 2.14. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$, form the product of AB . Is BA defined?

Solution. Since the number of columns of A = the number of rows of B (each being = 3).

$\therefore$ The product AB is defined and

$$= \begin{bmatrix} 0.1 + 1.(-1) + 2.2, & 0.(-2) + 1.0 + 2.(-1) \\ 1.1 + 2.(-1) + 3.2, & 1.(-2) + 2.0 + 3.(-1) \\ 2.1 + 3.(-1) + 4.2, & 2.(-2) + 3.0 + 4.(-1) \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 5 & -5 \\ 7 & -8 \end{bmatrix}$$

Again since the number of columns of $B \neq$ the number of rows of A .

$\therefore$ The product BA is not possible.

Example 2.15. If $A = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$, compute AB and BA and show that $AB \neq BA$.

Solution. Considering rows of A and columns of B , we have

$$AB = \begin{bmatrix} 1.2 + 3.1 + 0.(-1), & 1.3 + 3.2 + 0.1, & 1.4 + 3.3 + 0.2 \\ -1.2 + 2.1 + 1.(-1), & -1.3 + 2.2 + 1.1, & -1.4 + 2.3 + 1.2 \\ 0.2 + 0.1 + 2.(-1), & 0.3 + 0.2 + 2.1, & 0.4 + 0.3 + 2.2 \end{bmatrix} = \begin{bmatrix} 5 & 9 & 13 \\ -1 & 2 & 4 \\ -2 & 2 & 4 \end{bmatrix}$$

Again considering the rows of B and columns of A , we have

$$BA = \begin{bmatrix} 2.1 + 3.(-1) + 4.0, & 2.3 + 3.2 + 4.0, & 2.0 + 3.1 + 4.2 \\ 1.1 + 2.(-1) + 3.0, & 1.3 + 2.2 + 3.0, & 1.0 + 2.1 + 3.2 \\ -1.1 + 1.(-1) + 2.0, & -1.3 + 1.2 + 2.0, & -1.0 + 1.1 + 2.2 \end{bmatrix} = \begin{bmatrix} -1 & 12 & 11 \\ -1 & 7 & 8 \\ -2 & -1 & 5 \end{bmatrix}$$

Evidently $AB \neq BA$.

$$\begin{array}{r} 21 \\ 23 \\ \hline 44 \\ 314 \\ \hline 14 \\ 120 \\ \hline 2 \end{array}$$

Example 2.16. If $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 5 & 3 & 4 \end{bmatrix}$, find the matrix B such that $AB = \begin{bmatrix} 3 & 4 & 2 \\ 1 & 6 & 1 \\ 5 & 6 & 4 \end{bmatrix}$. (Mumbai, 2005)

Solution. Let $AB = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 5 & 3 & 4 \end{bmatrix} \begin{bmatrix} l & m & n \\ p & q & r \\ u & v & w \end{bmatrix}$

$$= \begin{bmatrix} 3l+2p+2u & 3m+2q+2v & 3n+2r+2w \\ l+3p+u & m+3q+v & n+3r+w \\ 5l+3p+4u & 5m+3q+4v & 5n+3r+4w \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 & 2 \\ 1 & 6 & 1 \\ 5 & 6 & 4 \end{bmatrix} \quad (\text{given})$$

Equating corresponding elements, we get

$$3l + 2p + 2u = 3, \quad l + 3p + u = 1, \quad 5l + 3p + 4u = 5 \quad \dots(i)$$

$$3m + 2q + 2v = 4, \quad m + 3q + v = 6, \quad 5m + 3q + 4v = 6 \quad \dots(ii)$$

$$3n + 2r + 2w = 2, \quad n + 3r + w = 1, \quad 5n + 3r + 4w = 4 \quad \dots(iii)$$

Solving the equations (i), we get $l = 1, p = 0, u = 0$

Similarly equations (ii) give $m = 0, q = 2, v = 0$

and equations (iii) give $n = 0, r = 0, w = 1$

Thus, $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Example 2.17. Prove that $A^3 - 4A^2 - 3A + 11I = 0$, where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$.

Solution. $A^2 = A \times A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+6+2 & 3+0+4 & 2-3+6 \\ 2+0-1 & 6+0-2 & 4+0-3 \\ 1+4+3 & 3+0+6 & 2-2+9 \end{bmatrix} = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix}$

$$A^3 = A^2 \times A = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 9+14+5 & 27+0+10 & 18-7+15 \\ 1+8+1 & 3+0+2 & 2-4+3 \\ 8+18+9 & 24+0+18 & 16-9+27 \end{bmatrix} = \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix}$$

$$A^3 - 4A^2 - 3A + 11I$$

$$= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - 4 \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} - 3 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 28-36-3+11 & 37-28-9+0 & 26-20-6+0 \\ 10-4-6-0 & 5-16+0+11 & 1-4+3+0 \\ 35-32-3+0 & 42-36-6+0 & 34-36-9+11 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0.$$

Example 2.18. By mathematical induction, prove that if

$$A = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}, \text{ then } A^n = \begin{bmatrix} 1+10n & -25n \\ 4n & 1-10n \end{bmatrix}$$

Solution. When $n = 1$, A^n gives $A^1 = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}$

Let us assume that the result is true for any positive integer k , so that

$$A^k = \begin{bmatrix} 1+10k & -25k \\ 4k & 1-10k \end{bmatrix}$$

$$A^{k+1} = A^k \cdot A^1 = \begin{bmatrix} 1+10k & -25k \\ 4k & 1-10k \end{bmatrix} \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}$$

$$= \begin{bmatrix} 11(1+10k) - 100k & -25(1+10k) + 225k \\ 44k + 4(1-10k) & -100k - 9(1-10k) \end{bmatrix}$$

$$= \begin{bmatrix} 1+10(k+1) & -25(k+1) \\ 4(k+1) & 1-10(k+1) \end{bmatrix}$$

This is true for $n = k + 1$

We have seen in (i) that the result is true for $n = 1$.

$\therefore$ It is true for $n = 1 + 1 = 2$

Similarly, it is true for $n = 2 + 1 = 3$ and so on.

Hence by mathematical induction, the result is true for all positive integers n .

Example 2.19. Prove that $(AB)C = A(BC)$, where A, B, C are matrices conformable for the products.
(J.N.T.U., 2002 S)

Solution. Let $A = [a_{ij}]$ be of order $m \times n$, $B = [b_{ij}]$ be of order $n \times p$ and $C = [c_{ij}]$ be of order of $p \times q$.

$$\text{Then } AB = [a_{ik}] [b_{kj}] = \sum_{k=1}^n a_{ik} b_{kj}$$

$$\therefore (AB)C = \left[\sum_{k=1}^n a_{ik} b_{kj} \right] \cdot [c_{ij}] = \left[\sum_{l=1}^p \left(\sum_{k=1}^n a_{ik} b_{kl} \right) c_{lj} \right] = \left[\sum_{k=1}^n \sum_{l=1}^p a_{ik} b_{kl} c_{lj} \right]$$

$$\text{Similarly, } BC = [b_{kl}] \cdot [c_{ij}] = \sum_{l=1}^p b_{kl} c_{lj}$$

$$\therefore A(BC) = [a_{ik}] \left[\sum_{l=1}^p b_{kl} c_{lj} \right] = \left[\sum_{k=1}^n a_{ik} \left(\sum_{l=1}^p b_{kl} c_{lj} \right) \right] = \left[\sum_{k=1}^n \left(\sum_{l=1}^p a_{ik} b_{kl} c_{lj} \right) \right]$$

$$\text{Hence } (AB)C = A(BC).$$

PROBLEMS 2.2

For what values of x , the matrix $\begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$ is singular?

Find the values of x, y, z and a which satisfy the matrix equation $\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4a-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2a \end{bmatrix}$.

Matrix A has x rows and $x+5$ columns. Matrix B has y rows and $11-y$ columns. Both AB and BA exist. Find

If $A + B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$ and $A - B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$, calculate the product AB .

If $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 0 & 1 & 2 \\ 3 & 1 & 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, find AB or BA , whichever exists.

If $A = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} -3 & 1 \\ 2 & 0 \end{bmatrix}$, verify that $(AB)C = A(BC)$ and $A(B+C) = AB + AC$.

Evaluate (i) $[x, y, z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; (ii) $\begin{bmatrix} 2 & 1 & -1 \\ 4 & -5 & 6 \\ -3 & 7 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & 1 \\ -6 & 4 \\ -2 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 5 & 3 \\ -2 & 1 \\ 3 \end{bmatrix}$; (iii) $\begin{bmatrix} 1 & 5 & 2 \\ -2 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} 4 & 5 & 2 \\ 5 & 2 & 1 \end{bmatrix}$

8. Prove that the product of two matrices

$$\begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \text{ and } \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

is a null matrix when θ and ϕ differ by an odd multiple of $\pi/2$.

(B.P.T.U., 2006)

9. If $A = \begin{bmatrix} 0 & -\tan \alpha/2 \\ \tan \alpha/2 & 0 \end{bmatrix}$, show that $I + A = (I - A)$ is a unit matrix of second order.

10. If $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$, find the value of $A^2 - 6A + 8I$, where I is a unit matrix of order 3, evaluate $A^2 - 3A + 9I$.

11. If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$, and I is the unit matrix of order 3, evaluate $A^2 - 3A + 9I$.

12. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 1 \\ 0 & 0 & 2 \\ 4 & -3 & 2 \end{bmatrix}$, verify the result $(A + B)^2 = A^2 + BA + AB + B^2$.

13. If $E = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$,

calculate the products EF and FE and show that $E^2F + FE^2 = E$.

14. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, show that $A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$, when n is a positive integer.

15. Factorize the matrix $A = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$ into the form LU , where L is lower triangular and U is upper triangular matrix.

2.6 RELATED MATRICES

(1) **Transpose of a matrix.** The matrix obtained from any given matrix A , by interchanging rows and columns is called the **transpose** of A and is denoted by A' .

$$\text{Thus the transposed matrix of } A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \text{ is } A' = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix}$$

Clearly, the transpose of an $m \times n$ matrix is an $n \times m$ matrix. Also the transpose of the transpose of a matrix coincides with itself, i.e., $(A')' = A$.

For a symmetric matrix, $A' = A$ and for a skew-symmetric matrix, $A' = -A$.

Obs. 1. The transpose of the product of the two matrices is the product of their transposes taken in the reverse order i.e., $(AB)' = B'A'$.

For, the element in the i th row and j th col. of $(AB)'$

= element in the j th row and i th col. of AB = inner product of j th row of A with i th col. of B

= inner product of j th col. of A' with i th row of $B' =$ element in the i th row and j th col. of $B'A'$

Hence $(AB)' = B'A'$.

Obs. 2. Every square matrix can be uniquely expressed as a sum of a symmetric and a skew-symmetric matrix.

(J.N.T.U., 2001)

Let A be the given square matrix, then $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$.

Let $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$

$\therefore B' = \left[\frac{1}{2}(A + A') \right]' = \frac{1}{2}[A' + (A')'] = \frac{1}{2}(A' + A) = B$, i.e., $B = \frac{1}{2}(A + A')$ is a symmetric matrix.

Again, $C' = \left[\frac{1}{2}(A - A') \right]' = \frac{1}{2}[A' - (A')'] = \frac{1}{2}(A' - A) = -C$, i.e., $C = \frac{1}{2}(A - A')$ is a skew-symmetric matrix.

Hence A can be expressed as the sum of a symmetric and a skew-symmetric matrix.

To prove the uniqueness, assume that P is a symmetric matrix and Q is a skew-symmetric matrix such that $A = P + Q$. Then $A' = (P + Q)' = P' + Q' = P - Q$.

Thus, $P = \frac{1}{2}(A + A')$ and $Q = \frac{1}{2}(A - A')$

which shows that there is one and only one way of expressing A as the sum of a symmetric and skew-symmetric matrix.

Example 2.20. Express the matrix A as the sum of a symmetric and a skew-symmetric matrix where

$$A = \begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix}$$

Solution. We have $A' = \begin{bmatrix} 4 & 1 & -5 \\ 2 & 3 & 0 \\ -3 & -6 & -7 \end{bmatrix}$

Then $A + A' = \begin{bmatrix} 8 & 3 & -8 \\ 3 & 6 & -6 \\ -8 & -6 & -14 \end{bmatrix}$ and $A - A' = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix}$

$$\therefore A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = \begin{bmatrix} 4 & 1.5 & -4 \\ 1.5 & 3 & -3 \\ -4 & -3 & -7 \end{bmatrix} + \begin{bmatrix} 0 & 0.5 & 1 \\ -0.5 & 0 & -3 \\ -1 & 3 & 0 \end{bmatrix}.$$

(2) Adjoint of a square matrix. The determinant of the square matrix

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \text{ is } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

The matrix formed by the cofactors of the elements in Δ is

$$\begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix}. \text{ Then the transpose of this matrix, i.e., } \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix},$$

is called the **adjoint** of the matrix A and is written as $\text{Adj. } A$.

Thus the **adjoint** of A is the transposed matrix of cofactors of A .

(3) Inverse of a matrix. If A be any matrix, then a matrix B if it exists, such that $AB = BA = I$, where I is the identity matrix, is called the **inverse** of A which is denoted by A^{-1} so that $AA^{-1} = I$.

Also
$$A^{-1} = \frac{\text{Adj. } A}{|A|}$$

For $A (\text{Adj. } A) = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$A \cdot \frac{\text{Adj. } A}{|A|} = I \quad [\because |A| \neq 0] \quad \text{or} \quad \frac{\text{Adj. } A}{|A|} \text{ is the inverse of } A.$$

Obs. 1. Inverse of a matrix, is unique.

If possible, let the two inverses of the matrix A be B and C ,

$$AB = BA = I$$

$$\text{and} \quad AC = CA = I$$

$$CAB = (CA)B = IB = B$$

$$\text{and} \quad CAB = C(AB) = CI = C$$

$$B = C.$$

$(AB)^{-1} = B^{-1}A^{-1}$

If A, B be two matrices, then the reciprocal of their product is $(AB)^{-1}$.

Clearly, $(AB) \cdot (B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$

$$= AIA^{-1} = AA^{-1} = I.$$

Similarly, $(B^{-1}A^{-1}) \cdot (AB) = I$

Hence $B^{-1}A^{-1}$ is the reciprocal of AB .

Obs. 3. Multiplication by an inverse matrix plays the same role in matrix algebra that division plays in ordinary algebra.

i.e., if

$$[A][B] = [C][D], \text{ then } [A]^{-1}[A][B] = [A^{-1}][C][D]$$

or

$$B = A^{-1}[C][D], \text{ i.e., } \frac{[C][D]}{[A]} = A^{-1}[C][D].$$

[by Associative law]

Example 2.21. Find the inverse of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

Solution. The determinant of the given matrix A is

$$\Delta = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \text{ (say)}$$

If $A_1, A_2, \dots$ be the cofactors of $a_1, a_2, \dots$ in Δ , then $A_1 = -24, A_2 = -8, A_3 = -12; B_1 = 10, B_2 = 2, B_3 = 6; C_1 = 2, C_2 = 2, C_3 = 2$.

Thus

$$\Delta = a_1A_1 + a_2A_2 + a_3A_3 = -8.$$

and

$$\text{adj } A = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} -24 & -8 & -12 \\ 10 & 2 & 6 \\ 2 & 2 & 2 \end{bmatrix}.$$

Hence the inverse of the given matrix A

$$= \frac{\text{adj } A}{\Delta} = \frac{1}{-8} \begin{bmatrix} -24 & -8 & -12 \\ 10 & 2 & 6 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 & \frac{3}{2} \\ -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

Note. For other methods see Examples 2.25; 2.28 and 2.46.

Example 2.22. Find the matrix A if $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix}$

Solution. If $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = B, \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = C$ and $\begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} = D$, then

$$BAC = D \text{ or } AC = B^{-1}D$$

$$A = B^{-1}DC^{-1}$$

$\therefore$

Now,

$$B^{-1} = \frac{\text{adj } B}{|B|} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

Similarly,

$$C^{-1} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

Hence,

$$\begin{aligned} A &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 14 & 8 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 24 & 13 \\ -34 & -18 \end{bmatrix}. \end{aligned}$$

PROBLEMS 2.3

1. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, verify that $AA' = I = A'A$, where I is the unit matrix.
2. Express each of the following matrices as the sum of a symmetric and a skew-symmetric matrix :
 (i) $\begin{bmatrix} 3 & -2 & 6 \\ 2 & 7 & -1 \\ 5 & 4 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} a & a & b \\ c & b & b \\ c & a & c \end{bmatrix}$
3. If A is a non-singular matrix of order n , prove that $A \operatorname{adj} A = |A| I$.
 Verify that $A (\operatorname{adj} A) = (\operatorname{adj} A) A = |A| I$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$.
 (Mumbai, 2006)
4. Find the inverse of the matrix (i) $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ (Mumbai, 2009) (ii) $\begin{bmatrix} 5 & -2 & 4 \\ -2 & 1 & 1 \\ 4 & 1 & 0 \end{bmatrix}$ (B.P.T.U., 2005)
5. If $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 5 & 3 & 4 \end{bmatrix}$, compute $\operatorname{adj} A$ and A^{-1} . Also find B such that $AB = \begin{bmatrix} 3 & 4 & 2 \\ 1 & 6 & 1 \\ 5 & 6 & 4 \end{bmatrix}$.
 (Mumbai, 2008)
6. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, (i) find A^{-1} ; (ii) show that $A^3 = A^{-1}$.
7. Find the inverse of the matrix
 $S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and if $A = \frac{1}{2} \begin{bmatrix} 4 & -1 & 1 \\ -2 & 3 & -1 \\ 2 & 1 & 5 \end{bmatrix}$,
 show that SAS^{-1} is a diagonal matrix $\operatorname{diag} (2, 3, 1)$.
 (Mumbai, 2007)
8. If $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$, prove that $A^{-1} = A'$.
9. Show that $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$.
10. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 4 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$, where A' is the transpose of A .
11. $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 9 & 3 \\ 1 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.
12. If A is a square matrix, show that (i) $A + A'$ is symmetric, and (ii) $A - A'$ is skew-symmetric.
 (P.T.U., 19)
13. If $D = \operatorname{diag} [d_1, d_2, d_3]$, $d_1, d_2, d_3 \neq 0$, prove that $D^{-1} = \operatorname{diag} [d_1^{-1}, d_2^{-1}, d_3^{-1}]$.
14. If A and B are square matrices of the same order and A is symmetrical, show that $B'AB$ is also symmetrical.
 [Hint. Show that $(B'AB)' = B'AB$]
15. If a non-singular matrix A is symmetric, show that A^{-1} is also symmetric.

2.7 (1) RANK OF A MATRIX

If we select any r rows and r columns from any matrix A , deleting all the other rows and columns, the determinant formed by these $r \times r$ elements is called the *minor of A of order r* . Clearly, there will be a number of different minors of the same order, got by deleting different rows and columns from the same matrix.

Def. A matrix is said to be of rank r when

- (i) it has at least one non-zero minor of order r ,
 and (ii) every minor of order higher than r vanishes.

Briefly, the rank of a matrix is the largest order of any non-vanishing minor of the matrix.

If a matrix has a non-zero minor of order r , its rank is $\geq r$.

If all minors of a matrix of order $r + 1$ are zero, its rank is $\leq r$.

The rank of a matrix A shall be denoted by $\rho(A)$.

(2) **Elementary transformation of a matrix.** The following operations, three of which refer to rows and three to columns are known as *elementary transformations*:

- I. The interchange of any two rows (columns).
- II. The multiplication of any row (column) by a non-zero number.
- III. The addition of a constant multiple of the elements of any row (column) to the corresponding elements of any other row (column).

Notation. The elementary row transformations will be denoted by the following symbols:

- (i) R_{ij} for the interchange of the i th and j th rows.
- (ii) kR_i for multiplication of the i th row by k .
- (iii) $R_i + pR_j$ for addition to the i th row, p times the j th row.

The corresponding column transformation will be denoted by writing C in place of R .

Elementary transformations do not change either the order or rank of a matrix. While the value of the minors may get changed by the transformation I and II, their zero or non-zero character remains unaffected.

(3) **Equivalent matrix.** Two matrices A and B are said to be *equivalent* if one can be obtained from the other by a sequence of elementary transformations. Two equivalent matrices have the same order and the same rank. The symbol $\sim$ is used for equivalence.

Example 2.23. Determine the rank of the following matrices:

$$(i) \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

(V.T.U., 2011)

Solution. (i) Operate $R_2 - R_1$ and $R_3 - 2R_1$ so that the given matrix

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} = A \text{ (say)}$$

Obviously, the 3rd order minor of A vanishes. Also its 2nd order minors formed by its 2nd and 3rd row are all zero. But another 2nd order minor is $\begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix} = -1 \neq 0$.

$\therefore \rho(A) = 2$. Hence the rank of the given matrix is 2.

(ii) Given matrix

$$\sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ 3 & 1 & -3 & -1 \\ 1 & 1 & -3 & -1 \end{bmatrix}$$

[Operating $C_3 - C_1, C_4 - C_1$]

$$\sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[Operating $R_3 - 3R_2, R_4 - R_2$]

$$\sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

[Operating $R_3 - R_1, R_4 - R_1$]

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = A \text{ (say)}$$

[Operating $C_3 + 3C_2, C_4 + C_2$]

Obviously, the 4th order minor of A is zero. Also every 3rd order minor of A is zero. But, of all the 2

minors, only $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1 \neq 0$. $\therefore \rho(A) = 2$.

Hence the rank of the given matrix is 2.

(4) **Elementary matrices.** An elementary matrix is that, which is obtained from a unit matrix, by subjecting any of the elementary transformations.

Examples of elementary matrices obtained from

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ are } R_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = C_{23}; kR_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}; R_1 + pR_2 = \begin{bmatrix} 1 & p & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(5) **Theorem.** Elementary row (column) transformations of a matrix A can be obtained by pre-multiplying (post-multiplying) A by the corresponding elementary matrices.

Consider the matrix $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

Then $R_{23} \times A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{bmatrix}$

So a pre-multiplication by R_{23} has interchanged the 2nd and 3rd rows of A . Similarly, pre-multiplication by kR_2 will multiply the 2nd row of A by k and pre-multiplication by $R_1 + pR_2$ will result in the addition of p times the 2nd row of A to its 1st row.

Thus the pre-multiplication of A by elementary matrices results in the corresponding elementary row transformations of A . It can easily be seen that post multiplication will perform the elementary column transformations.

(6) **Gauss-Jordan method of finding the inverse***. Those elementary row transformations which reduce a given square matrix A to the unit matrix, when applied to unit matrix I give the inverse of A .

Let the successive row transformations which reduce A to I result from pre-multiplication by the elementary matrices $R_1, R_2, \dots, R_i$ so that

$$R_i R_{i-1} \dots R_2 R_1 A = I$$

$$R_i R_{i-1} \dots R_2 R_1 A A^{-1} = I A^{-1}$$

$$R_i R_{i-1} \dots R_2 R_1 I = A^{-1}$$

$$[\because AA^{-1} = I]$$

Hence the result.

Working rule to evaluate A^{-1} . Write the two matrices A and I side by side. Then perform the same row transformations on both. As soon as A is reduced to I , the other matrix represents A^{-1} .

Example 2.24. Using the Gauss-Jordan method, find the inverse of the matrix

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

(Kurukshetra, 2006)

Solution. Writing the same matrix side by side with the unit matrix of order 3, we have

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & -3 & 0 & 1 & 0 \\ -2 & -4 & -4 & 0 & 0 & 1 \end{array} \right] \quad (R_2 - R_1)$$

(Operate $R_2 - R_1$ and $R_3 + 2R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{array} \right]$$

(Operate $\frac{1}{2}R_2$ and $\frac{1}{2}R_3$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & 1 & 0 & \frac{1}{2} \end{array} \right]$$

(Operate $R_1 - R_2$ and $R_3 + R_2$)

named after the great German mathematician Carl Friedrich Gauss (1777–1855) who made his first great discovery at Göttingen. His important contributions are to algebra, number theory, mechanics, complex analysis, differential geometry, non-Euclidean geometry, numerical analysis, astronomy and electromagnetism. He was the director of the observatory at Göttingen in 1807.

named after another German mathematician and geodesist Wilhelm Jordan (1842–1899).

$$\sim \begin{bmatrix} 1 & 0 & 6 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -2 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\left[\text{Operate } R_1 + 3R_3, R_2 - \frac{3}{2}R_3 \text{ and } \left(-\frac{1}{2}\right)R_2 \right]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 3 & 1 & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & \frac{3}{2} \\ -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

[cf. Example 2.21]

Hence the inverse of the given matrix is

(7) **Normal form of a matrix.** Every non-zero matrix A of rank r , can be reduced by a sequence of elementary transformations, to the form

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ called the normal form of } A. \quad \dots(i)$$

Cor. 1. The rank of a matrix A is r if and only if it can be reduced to the normal form (i).

Cor. 2. Since each elementary transformation can be affected by pre-multiplication or post-multiplication with a suitable elementary matrix and each elementary matrix is non-singular, therefore, we have the following result :

Corresponding to every matrix A of rank r , there exist non-singular matrices P and Q such that PAQ equals (i).

If A be a $m \times n$ matrix, then P and Q are square matrices of orders m and n respectively.

Example 2.25. Reduce the following matrix into its normal form and hence find its rank.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(U.P.T.U., 2005)

Solution.

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

[By R_{12}]

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

[By $R_2 - 2R_1, R_3 - 3R_1, R_4 - 6R_1$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

[By $C_2 + C_1, C_3 + 2C_1, C_4 + 4C_1$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $R_4 - R_2 - R_3$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $R_2 - R_1$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $R_3 - 4R_2$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $C_3 + 6C_2, C_4 + 3C_2$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $\frac{1}{33}$]

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[By $C_4 -$]

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence $\rho(A) = 3$.

Example 2.26. For the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$,

find non-singular matrices P and Q such that PAQ is in the normal form. Hence find the rank of A .
(Kurukshetra)

Solution. We write $A = IAI$, i.e., $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

We shall affect every elementary row (column) transformation of the product by subjecting the (post-factor) of A to the same.

Operate $C_2 - C_1, C_3 - 2C_1$, $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Operate $R_2 - R_1$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Operate $C_3 - C_2$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Operate $R_3 + R_2$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

rank

ch is of the normal form $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$

Hence, $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$ and $\rho(A) = 2$.

PROBLEMS 2.1

Determine the rank of the following matrices (1-4):

1. $\begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix}$

(P.T.U., 2005)

2. $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$

(W.B.T.U., 2005)

3. $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$

(Kottayam, 2005)

4. $\begin{bmatrix} 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 9 \\ 11 & 12 & 13 & 14 \\ 16 & 17 & 18 & 19 \end{bmatrix}$

(Rohtak, 2004)

5. $\begin{bmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{bmatrix}$

(Bhopal, 2008)

6. Determine the values of p such that the rank of $A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ p & 2 & 2 & 2 \\ 4 & 9 & p & 3 \end{bmatrix}$ is 3.

(Mumbai, 2007)

7. Use Gauss-Jordan method to find the inverse of the following matrices:

(i) $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$

(ii) $\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

(Mumbai, 2008)

(iii) $\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ (B.P.T.U., 2006)

(iv) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

(Kurukshetra, 2006)

8. Find the non-singular matrices P and Q such that $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{bmatrix}$ is reduced to normal form. Also find its rank.

(S.V.T.U., 2009 ; Mumbai, 200)

9. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1} . Also find two non-singular matrices P and Q such that $PAQ = I$, where I is the unit matrix and verify that $A^{-1} = QP$.

10. Find non-singular matrices P and Q such that PAQ is in the normal form for the matrices:

(i) $A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ (Rohtak, 2004)

(ii) $A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$

11. Reduce each of the following matrices to normal form and hence find their ranks:

(i) $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$ (Kurukshetra, 2005)

(ii) $A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$

(Bhopal 2004)

(iii) $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ (Mumbai, 2008)

(iv) $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$

(U.T.U., 2004)

2.8 PARTITION METHOD OF FINDING THE INVERSE

According to this method of finding the inverse, if the inverse of a matrix A_n of order n is known, then the inverse of the matrix A_{n+1} can easily be obtained by adding $(n+1)$ th row and $(n+1)$ th column to A_n .

Let
$$A = \begin{bmatrix} A_1 & : & A_2 \\ \dots & : & \dots \\ A_3' & : & \alpha \end{bmatrix} \text{ and } A^{-1} = \begin{bmatrix} X_1 & : & X_2 \\ \dots & : & \dots \\ X_3' & : & x \end{bmatrix}$$

where A_2, X_2 are column vectors and A_3', X_3' are row vectors (being transposes of column vectors A_3, X_3) and α, x are ordinary numbers. We also assume that A_1^{-1} is known.

Then, $AA^{-1} = I_{n+1}$, i.e.,
$$\begin{bmatrix} A_1 & : & A_2 \\ \dots & : & \dots \\ A_3' & : & \alpha \end{bmatrix} \begin{bmatrix} X_1 & : & X_2 \\ \dots & : & \dots \\ X_3' & : & x \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ 0 & 1 \end{bmatrix}$$

gives

$$\begin{aligned} A_1 X_1 + A_2 X_3' &= I_n \\ A_1 X_2 + A_2 x &= 0 \\ A_3' X_1 + \alpha X_3' &= 0 \\ A_3' X_2 + \alpha x &= 1 \end{aligned}$$

From (ii), $X_2 = -A_1^{-1} A_2 x$ and using this, (iv) gives $x = (\alpha - A_3' A_1^{-1} A_2)^{-1}$

Hence x and then X_2 are given.

Also from (i), $X_1 = A_1^{-1} (I_n - A_2 X_3')$

and using this, (iii) gives $X_3' = -A_3' A_1^{-1} (\alpha - A_3' A_1^{-1} A_2)^{-1} = -A_3' A_1^{-1} x$

Then X_1 is determined and hence A^{-1} is computed.

Obs. This is also known as the 'Escalator method'. For evaluation of A^{-1} we only need to determine two inverse matrices A_1^{-1} and $(\alpha - A_3' A_1^{-1} A_2)^{-1}$.

Example 2.27. Using the partition method, find the inverse of $\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$.

Solution. Let

$$A = \begin{bmatrix} 1 & 1 & : & 1 \\ 4 & 3 & : & -1 \\ \dots & \dots & : & \dots \\ 3 & 5 & : & 3 \end{bmatrix} = \begin{bmatrix} A_1 & : & A_2 \\ \dots & : & \dots \\ A_3' & : & \alpha \end{bmatrix}$$

$$A_1^{-1} = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}^{-1} = -\begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

Let

$$A^{-1} = \begin{bmatrix} X_1 & : & X_2 \\ \dots & : & \dots \\ X_3' & : & x \end{bmatrix} \text{ so that } AA^{-1} = I.$$

$$\alpha - A_3' A_1^{-1} A_2 = 3 + [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -10$$

$$x = (\alpha - A_3' A_1^{-1} A_2)^{-1} = -\frac{1}{10}$$

Also,

$$X_2 = -A_1^{-1} A_2 x = \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(-\frac{1}{10}\right) = -\frac{1}{10} \begin{bmatrix} 4 \\ -5 \end{bmatrix}$$

Then

$$X_3' = -A_3' A_1^{-1} x = [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \left(-\frac{1}{10}\right) = -\frac{1}{10} [-11 \ 2]$$

Finally,

$$X_1 = A_1^{-1} (I - A_2 X_3') = -\begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} [-11 \ 2]$$

$$= \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} -44 & 8 \\ 55 & -10 \end{bmatrix} = \begin{bmatrix} 1.4 & 0.2 \\ -1.5 & 0 \end{bmatrix}$$

Hence

$$A^{-1} = \begin{bmatrix} 1.4 & 0.2 & -0.4 \\ -1.5 & 0 & 0.5 \\ 1.1 & -0.2 & -0.1 \end{bmatrix}$$

Example 2.28. If A and C are non-singular matrices, then show that $\begin{bmatrix} A & O \\ B & C \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & 0 \\ -C^{-1}BA^{-1} & C^{-1} \end{bmatrix}$

Hence find inverse of $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 1 & 0 & 3 \end{bmatrix}$.

(Mumbai, 2005)

Solution. Let the given matrix be $M = \begin{bmatrix} A & O \\ B & C \end{bmatrix}$ and its inverse be $M^{-1} = \begin{bmatrix} P & Q \\ R & S \end{bmatrix}$ both in the partitioned form where A, B, C, P, Q, R, S are all matrices.

$$MM^{-1} = \begin{bmatrix} A & O \\ B & C \end{bmatrix} \begin{bmatrix} P & Q \\ R & S \end{bmatrix} = I$$

or

$$\begin{bmatrix} AP + OR & AQ + OS \\ BP + CR & BQ + CS \end{bmatrix} = \begin{bmatrix} I & O \\ O & I \end{bmatrix}$$

$\therefore$ Equating corresponding elements, we have

$$AP + OR = I, AQ + OS = 0, BP + CR = 0, BQ + CS = I.$$

Second relation gives $AQ = 0$, i.e., $Q = 0$ as A is non-singular.

First relation gives $AP = I$, i.e., $P = A^{-1}$.

From third equation, $BP + CR = 0$, i.e., $CR = -BP = -BA^{-1}$

$\therefore C^{-1}CR = -C^{-1}BA^{-1}$ or $IR = -C^{-1}BA^{-1}$ or $R = -C^{-1}BA^{-1}$

From fourth equation, $BQ + CS = I$, or $CS = I$ or $S = C^{-1}$

Hence

$$M^{-1} = \begin{bmatrix} A^{-1} & 0 \\ -C^{-1}BA^{-1} & C^{-1} \end{bmatrix}$$

(ii) Let

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} A & O \\ B & C \end{bmatrix}$$

Whence

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, C^{-1} = \frac{1}{12} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$$

$$-C^{-1}(BA^{-1}) = -\frac{1}{12} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \left\{ \frac{1}{2} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$= -\frac{1}{24} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix} = -\frac{1}{24} \begin{bmatrix} 18 & 0 \\ 0 & 4 \end{bmatrix}$$

$$M^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ -3/4 & 0 & 1/4 & 0 \\ 0 & -1/6 & 0 & 1/3 \end{bmatrix}$$

nce,

PROBLEMS 2.5

Find the inverse of each of the following matrices using the partition method :

1. $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

(Nagpur, 1997)

2. $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 3 & -1 & 10 & 2 \\ 5 & 1 & 20 & 3 \\ 9 & 7 & 39 & 4 \\ 1 & -2 & 2 & 1 \end{bmatrix}$

2.9 SOLUTION OF LINEAR SYSTEM OF EQUATIONS

(1) Method of determinants—Cramer's* rule

Consider the equations $\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$

...(i)

If the determinant of coefficient be $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

then

$$x\Delta = \begin{vmatrix} xa_1 & b_1 & c_1 \\ xa_2 & b_2 & c_2 \\ xa_3 & b_3 & c_3 \end{vmatrix} \quad [\text{Operate } C_1 + yC_2 + zC_3]$$

$$= \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

[By (

Thus

$$x = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}} \quad \text{provided } \Delta \neq 0.$$

Similarly,

$$y = \frac{\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

$$z = \frac{\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

Equation (ii), (iii) and (iv) giving the values of x , y , z constitute the **Cramer's rule**, which reduces the solution of the linear equations (i) to a problem in evaluation of determinants.

(2) Matrix inversion method

If $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

the equations (i) are equivalent to the matrix equation $AX = D$ where A is the coefficient matrix.

Multiplying both sides of (v) by the reciprocal matrix A^{-1} , we get

$$A^{-1}AX = A^{-1}D \quad \text{or} \quad IX = A^{-1}D$$

[$\therefore$

iel Cramer (1704–1752), a Swiss mathematician.

or

$$X = A^{-1}D \quad \text{i.e.,} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \times \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \quad \dots (vi)$$

where A_1, B_1 etc. are the cofactors of a_1, b_1 etc. in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} (\Delta \neq 0)$

Hence equating the values of x, y, z to the corresponding elements in the product on the right side of (vi), we get the desired solutions.

Obs. When A is a singular matrix, i.e., $\Delta = 0$, the above methods fail. These also fail when the number of equations and the number of unknowns are unequal. Matrices can, however, be usefully applied to deal with such equations as will be seen in § 2.10(2).

Example 2.29. Solve the equations $3x + y + 2z = 3$, $2x - 3y - z = -3$, $x + 2y + z = 4$ by (i) determinants (ii) matrices.

Solution. (i) By determinants :

Here $\Delta = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 3(-3 + 2) - 2(1 - 4) + (-1 + 6) = 8$ [Expanding by C_1]

$$x = \frac{1}{\Delta} \begin{vmatrix} 3 & 1 & 2 \\ -3 & -3 & -1 \\ 4 & 2 & 1 \end{vmatrix} \quad [\text{Expand by } C_1]$$

$$= \frac{1}{8} [3(-3 + 2) + 3(1 - 4) + 4(-1 + 6)] = 1$$

Similarly, $y = \frac{1}{\Delta} \begin{vmatrix} 3 & 3 & 2 \\ 2 & -3 & -1 \\ 1 & 4 & 1 \end{vmatrix} = 2$ and $z = \frac{1}{\Delta} \begin{vmatrix} 3 & 1 & 3 \\ 2 & -3 & -3 \\ 1 & 2 & 4 \end{vmatrix} = -1$

Hence $x = 1, y = 2, z = -1$.

Note. The use of Cramer's rule involves a lot of labour when the number of equations exceeds four. In such and other cases, the numerical methods given in § 28.4 to 28.6 are preferable.

(ii) By matrices :

Here $\Delta = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad (\text{say}).$

Then $A_1 = -1, A_2 = 3, A_3 = 5; B_1 = -3, B_2 = 1, B_3 = 7; C_1 = 7, C_2 = -5, C_3 = -11$.

Also $\Delta = a_1 A_1 + a_2 A_2 + a_3 A_3 = 8.$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \times \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} -1 & 3 & 5 \\ -3 & 1 & 7 \\ 7 & -5 & -11 \end{bmatrix} \times \begin{bmatrix} 3 \\ -3 \\ 4 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} -3 - 9 + 20 \\ -9 - 3 + 28 \\ 21 + 15 - 44 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

∴ $x = 1, y = 2, z = -1$.

Example 2.30. Solve the equations $x_1 - x_2 + x_3 + x_4 = 2$; $x_1 + x_2 - x_3 + x_4 = -4$; $x_1 + x_2 + x_3 - x_4 = 4$; $x_1 + x_4 = 0$, by finding the inverse by elementary row operations.

Ans. Given system can be written as $AX = B$, where

$$A = \begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, B = \begin{bmatrix} 2 \\ -4 \\ 4 \\ 0 \end{bmatrix}$$

To find A^{-1} , we write

$$\begin{aligned}
 [A : I] &= \begin{bmatrix} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & -1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_2 - R_1 \\ R_3 + R_1 \\ R_4 + R_1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 2 & -2 & 0 & -1 & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\ 2 & 0 & 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2}R_2 \\ \frac{1}{2}R_3 \\ \frac{1}{2}R_4 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & -1/2 & 1/2 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1/2 & 0 & 1/2 & 0 \\ 1 & 0 & 1 & 1 & 1/2 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} R_3 - R_2 \\ R_4 - R_3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 1 & 1/2 & 1/2 & 0 & 0 \\ 0 & 1 & -1 & 0 & -1/2 & 1/2 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} R_3 - R_4 \\ R_2 + R_3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1/2 & 1/2 & +1/2 & -1/2 \\ 1 & 1 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 1 & 0 & 1 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} R_2 - R_1 \\ R_3 - R_1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1/2 & 1/2 & 1/2 & -1/2 \\ 0 & 1 & 0 & 0 & -1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1/2 & 1/2 \end{bmatrix}
 \end{aligned}$$

Thus,

$$A^{-1} = \begin{bmatrix} 1/2 & 1/2 & 1/2 & -1/2 \\ -1/2 & 0 & 0 & 1/2 \\ 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & -1/2 & 1/2 \end{bmatrix}$$

Hence,

$$X = A^{-1}B = \begin{bmatrix} 1/2 & 1/2 & 1/2 & -1/2 \\ -1/2 & 0 & 0 & 1/2 \\ 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ -4 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}$$

i.e.,

$$x_1 = 1, x_2 = -1, x_3 = 2, x_4 = -2.$$

$$\begin{aligned}
 AX &= B \\
 A^{-1}AX &= A^{-1}B \\
 X &= A^{-1}B
 \end{aligned}$$

PROBLEMS 2.6

Solve the following equations with the help of determinants (1 to 4) :

- $x + y + z = 4$; $x - y + z = 0$; $2x + y + z = 5$.
- $x + 3y + 6z = 2$; $3x - y + 4z = 9$; $x - 4y + 2z = 7$.
- $x + y + z = 6.6$; $x - y + z = 2.2$; $x + 2y + 3z = 15.2$.
- $x^2 z^3 / y = e^8$; $y^2 z / x = e^4$; $x^3 y / z^4 = 1$.
- $2vw - wu + uv = 3uvw$; $3vw + 2wu + 4uv = 19uvw$; $6vw + 7wu - uv = 17uvw$.

(Osmania, 20)

Solve the following system of equations by matrix method (6 to 8) :

- $x_1 + x_2 + x_3 = 1$, $x_1 + 2x_2 + 3x_3 = 6$, $x_1 + 3x_2 + 4x_3 = 6$.
- $x + y + z = 3$; $x + 2y + 3z = 4$; $x + 4y + 9z = 6$.
- $2x - 3y + 4z = -4$, $x + z = 0$, $-y + 4z = 2$.
- $2x - y + 3z = 8$; $x - 2y - z = -4$; $3x + y - 4z = 0$.
- $2x_1 + x_2 + 2x_3 + x_4 = 6$, $4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$, $6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$, $2x_1 + 2x_2 - x_3 + x_4 = 10$.

(P.T.U., 2)

(Bhopal, 2)

(W.B.T.U., 1)

(Mumbai, 1)

(U.P.T.U., 1)

11. By finding A^{-1} , solve the linear equation $AX = B$, where $A = \begin{bmatrix} 3 & 4 & 5 \\ 1 & 2 & 0 \\ 5 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$.
12. In a given electrical network, the equations for the currents i_1, i_2, i_3 are
 $3i_1 + i_2 + i_3 = 8$; $2i_1 - 3i_2 - 2i_3 = -5$; $7i_1 + 2i_2 - 5i_3 = 0$.
 Calculate i_1 and i_3 by Cramer's rule.
13. Using the loop current method on a circuit, the following equations are obtained:
 $7i_1 - 4i_2 = 12$, $-4i_1 + 12i_2 - 6i_3 = 0$, $-6i_2 + 14i_3 = 0$.
 By matrix method, solve for i_1, i_2 and i_3 .
14. Solve the following equations by calculating the inverse by elementary row operations:
 $2x_1 + 2x_2 + 2x_3 - 3x_4 = 2$; $3x_1 + 6x_2 - 2x_3 + x_4 = 8$; $x_1 + x_2 - 3x_3 - 4x_4 = -1$; $2x_1 + x_2 + 5x_3 + x_4 = 5$.

2.10 (1) CONSISTENCY OF LINEAR SYSTEM OF EQUATIONS

Consider the system of m linear equations

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= k_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= k_2 \\ \dots &\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= k_m \end{aligned} \right\} \dots (i)$$

containing the n unknowns $x_1, x_2, \dots, x_n$. To determine whether the equations (i) are consistent (i.e., possess a solution) or not, we consider the ranks of the matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad \text{and} \quad K = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & k_1 \\ a_{21} & a_{22} & \dots & a_{2n} & k_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & k_m \end{bmatrix}$$

A is the co-efficient matrix and K is called the augmented matrix of the equations (i).

(2) Rouché's theorem. The system of equations (i) is consistent if and only if the coefficient matrix A and the augmented matrix K are of the same rank otherwise the system is inconsistent.

Proof. We consider the following two possible cases:

I. Rank of $A = \text{rank of } K = r$ ($r \leq$ the smaller of the numbers m and n). The equations (i) can, be suitable row operations, be reduced to

$$\left. \begin{aligned} b_{11}x_1 + b_{12}x_2 + \dots + b_{1n}x_n &= l_1 \\ 0.x_1 + b_{22}x_2 + \dots + b_{2n}x_n &= l_2 \\ \dots &\dots \\ 0.x_1 + 0.x_2 + \dots + b_{rn}x_n &= l_r \end{aligned} \right\} \dots (ii)$$

the remaining $m - r$ equations being all of the form $0.x_1 + 0.x_2 + \dots + 0.x_n = 0$.

The equations (ii) will have a solution, though $n - r$ of the unknowns may be chosen arbitrarily. The solution, will be unique only when $r = n$. Hence the equations (i) are consistent.

II. Rank of A (i.e., r) $<$ rank of K . In particular, let the rank of K be $r + 1$. In this case, the equations (i) will, by suitable row operations, to

$$\begin{aligned} b_{11}x_1 + b_{12}x_2 + \dots + b_{1n}x_n &= l_1, \\ 0.x_1 + b_{22}x_2 + \dots + b_{2n}x_n &= l_2, \\ \dots &\dots \\ 0.x_1 + 0.x_2 + \dots + b_{rn}x_n &= l_r, \\ 0.x_1 + 0.x_2 + \dots + 0.x_n &= l_{r+1}, \end{aligned}$$

remaining $m - (r + 1)$ equations are of the form $0.x_1 + 0.x_2 + \dots + 0.x_n = 0$.
 rly, the $(r + 1)$ th equation cannot be satisfied by any set of values for the unknowns. Hence the equations (i) are inconsistent.

Procedure to test the consistency of a system of equations in n unknowns:
 Reduce the coefficient matrix A and the augmented matrix K , by reducing A to the triangular form by elementary row operations. Let the rank of A be r and that of K be r' .

- (i) If $r \neq r'$, the equations are inconsistent, i.e., there is no solution.
 (ii) If $r = r' = n$, the equations are consistent and there is a unique solution.
 (iii) If $r = r' < n$, the equations are consistent and there are infinite number of solutions. Giving arbitrary values to $n - r$ of the unknowns, we may express the other r unknowns in terms of these.]

Example 2.31. Test for consistency and solve

$$5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5.$$

(Bhopal, 2008 ; J.N.T.U., 2005 ; P.T.U., 2005)

Solution. We have

$$\begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

Operate $3R_1, 5R_2,$

$$\begin{bmatrix} 15 & 9 & 21 \\ 15 & 130 & 10 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 45 \\ 5 \end{bmatrix}$$

Operate $R_2 - R_1,$

$$\begin{bmatrix} 15 & 9 & 21 \\ 0 & 121 & -11 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 33 \\ 5 \end{bmatrix}$$

Operate $\frac{7}{3}R_1, 5R_3, \frac{1}{11}R_2,$

$$\begin{bmatrix} 35 & 21 & 49 \\ 0 & 11 & -1 \\ 35 & 10 & 50 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 28 \\ 3 \\ 25 \end{bmatrix}$$

Operate $R_3 - R_1 + R_2, \frac{1}{7}R_1,$

$$\begin{bmatrix} 5 & 3 & 7 \\ 0 & 11 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

The ranks of coefficient matrix and augmented matrix for the last set of equations, are both 2. Hence the equations are consistent. Also the given system is equivalent to

$$5x + 3y + 7z = 4, 11y - z = 3, \therefore y = \frac{3}{11} + \frac{z}{11} \text{ and } x = \frac{7}{11} - \frac{16}{11}z$$

where z is a parameter.

Hence $x = \frac{7}{11}, y = \frac{3}{11}$ and $z = 0$, is a particular solution.

Obs. In the above solution, the coefficient matrix is reduced to an upper triangular matrix by row-transformations

Example 2.32. Investigate the values of λ and μ so that the equations

$$2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu,$$

have (i) no solution, (ii) a unique solution and (iii) an infinite number of solutions.

(Mumbai, 2007 ; V.T.U., 2006)

Solution. We have
$$\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

The system admits of unique solution if, and only if, the coefficient matrix is of rank 3. This requires

$$\begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{vmatrix} = 15(5 - \lambda) \neq 0$$

Thus for a unique solution $\lambda \neq 5$ and μ may have any value. If $\lambda = 5$, the system will have no solution those values of μ for which the matrices

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & 5 \end{bmatrix} \text{ and } K = \begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & 5 & \mu \end{bmatrix}$$

Rank of coefficient and augmented matrix.
for the last set of eqⁿ is
= 2
2 row

15(5-λ) ≠ 0

15(5-λ) ≠ 0

15(5-λ) ≠ 0

15(5-λ) ≠ 0

If $\lambda = 5$ and $\mu = 9$, the system will have an infinite number of solutions.

Example 2.33. Test for consistency the following equations and solve them if consistent : $x - 2y + 3z = 2$, $2x + y + z + t = -4$; $4x - 3y + z + 7t = 8$.
(Mumbai, 2008)

Solution. Given equation can be written as

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & 1 & 1 & 1 \\ 4 & -3 & 1 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 8 \end{bmatrix}$$

Operate $R_2 - 2R_1$, $R_3 - 4R_1$,

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 0 & 5 & 1 & -5 \\ 0 & 5 & 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

Operate $R_3 - R_2$,

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 0 & 5 & 1 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

Clearly, rank of the coefficient matrix is 2 and the rank of augmented matrix is also 2. Hence the given equations are consistent. But the rank $2 < 4$, the number of unknowns.

$\therefore$ The number of parameters is $4 - 2 = 2$

Thus the equations have doubly infinite solutions. Now putting $t = k_1$ and $z = k_2$ in

$$x - 2y + 3t = 2 \quad \text{and} \quad 5y + z - 5t = 0,$$

we get $x - 2y + 3k_1 = 2$ and $5y + k_2 - 5k_1 = 0$

Hence

$$y = k_1 - k_2/5.$$

$$x = 2 + 2y - 3k_1$$

$$= 2 + 2(k_1 - k_2/5) - 3k_1$$

$$= 2 - k_1 - \frac{2}{5}k_2$$

(3) **System of linear homogeneous equations.** Consider the homogeneous linear equations

$$\left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0 \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0 \end{array} \right\} \dots(iii)$$

Find the rank r of the coefficient matrix A by reducing it to the triangular form by elementary row operations.

I. If $r = n$, the equations (iii) have only a trivial zero solution

$$x_1 = x_2 = \dots = x_n = 0$$

If $r < n$, the equation (iii) have $(n - r)$ linearly independent solutions. The number of solutions is

The number of linearly independent solutions is $(n - r)$ means, if arbitrary values are assigned to $(n - r)$ of variables, the values of the remaining variables can be uniquely found.

Thus the equations (iii) will have an infinite number of solutions.

II. When $m < n$ (i.e., the number of equations is less than the number of variables), the solution is always $x_1 = x_2 = \dots = x_n = 0$. The number of solutions is infinite.

II. When $m = n$ (i.e., the number of equations = the number of variables), the necessary and sufficient condition for solutions other than $x_1 = x_2 = \dots = x_n = 0$, is that the determinant of the coefficient matrix is zero. In this case the equations are said to be consistent and such a solution is called non-trivial solution. The determinant is called the **eliminant** of the equations.

Example 2.34. Solve the equations

- (i) $x + 2y + 3z = 0$, $3x + 4y + 4z = 0$, $7x + 10y + 12z = 0$
 (ii) $4x + 2y + z + 3w = 0$, $6x + 3y + 4z + 7w = 0$, $2x + y + w = 0$.

Solution. (i) Rank of the coefficient matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 7 & 10 & 12 \end{bmatrix}$$

[Operating $R_3 - 3R_1$]

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

[Operating $R_3 - 7R_1 - 2R_2$]

is 3 which = the number of variables (i.e., $r = n$)

$\therefore$ The equations have only a trivial solution : $x = y = z = 0$.

(ii) Rank of the coefficient matrix

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 5/2 & 5/2 \\ 0 & 0 & -1/2 & -1/2 \end{bmatrix}$$

[Operating $R_2 - \frac{3}{2}R_1$, $R_3 - \frac{1}{2}R_1$]

$$\sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 5/2 & 5/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[Operating $R_2 + \frac{1}{5}R_2$]

is 2 which < the number of variable (i.e., $r < n$)

$\therefore$ Number of independent solutions = $4 - 2 = 2$. Given system is equivalent to

$$4x + 2y + z + 3w = 0, z + w = 0.$$

$\therefore$ We have $z = -w$ and $y = -2x - w$

which give an infinite number of non-trivial solutions, x and w being the parameters.

Example 2.35. Find the values of k for which the system of equations $(3k - 8)x + 3y + 3z = 0$, $3x + (3k - 8)y + 3z = 0$, $3x + 3y + (3k - 8)z = 0$ has a non-trivial solution. (U.P.T.U., 2006)

Solution. For the given system of equations to have a non-trivial solution, the determinant of the coefficient matrix should be zero.

$$\text{i.e., } \begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} 3k-2 & 3 & 3 \\ 3k-2 & 3k-8 & 3 \\ 3k-2 & 3 & 3k-8 \end{vmatrix} = 0 \quad [\text{Operating } C_1 + (C_2 + C_3)]$$

$$\text{or } (3k-2) \begin{vmatrix} 1 & 3 & 3 \\ 1 & 3k-8 & 3 \\ 1 & 3 & 3k-8 \end{vmatrix} = 0 \quad \text{or} \quad (3k-2) \begin{vmatrix} 1 & 3 & 3 \\ 0 & 3k-11 & 0 \\ 0 & 0 & 3k-11 \end{vmatrix} = 0 \quad [\text{Operating } R_2 - R_1, R_3 - R_1]$$

$$\text{or } (3k-2)(3k-11)^2 = 0 \text{ whence } k = 2/3, 11/3, 11/3.$$

Example 2.36. If the following system has non-trivial solution, prove that $a + b + c = 0$ or $a = b = c$.
 $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$. (Mumbai, 2006)

Solution. For the given system of equations to have non-trivial solution, the determinant of the coefficient matrix is zero.

$$\text{i.e., } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \quad [\text{Operating } R_1 + R_2 + R_3]$$

$$(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \quad \text{or} \quad (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ b & c-b & a-b \\ c & a-c & b-c \end{vmatrix} = 0 \quad [\text{Operating } C_2 - C_1, C_3 - C_1]$$

or

$$(a + b + c) [(c - b)(b - c) - (a - c)(a - b)] = 0$$

or

$$(a + b + c)(-a^2 - b^2 - c^2 + ab + bc + ca) = 0$$

i.e.,

$$a + b + c = 0 \quad \text{or} \quad a^2 + b^2 + c^2 - ab - bc - ca = 0$$

or

$$a + b + c = 0 \quad \text{or} \quad \frac{1}{2} [(a - b)^2 + (b - c)^2 + (c - a)^2] = 0$$

or

$$a + b + c = 0; a = b, b = c, c = a.$$

Hence the given system has a non-trivial solution if $a + b + c = 0$ or $a = b = c$.

Example 2.37. Find the values of λ for which the equations

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

are consistent, and find the ratios of $x : y : z$ when λ has the smallest of these values. What happens when λ has the greatest of these values. (Kurukshetra, 2006 ; Delhi, 2002)

Solution. The given equations will be consistent, if

$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3(\lambda - 1) \end{vmatrix} = 0$$

[Operate $R_2 - R_1$]

if,

$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ 0 & \lambda - 3 & 3 - \lambda \\ 2 & 3\lambda + 1 & 3(\lambda - 1) \end{vmatrix} = 0$$

[Operate $C_3 + C_2$]

if,

$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 5\lambda + 1 \\ 0 & \lambda - 3 & 0 \\ 2 & 3\lambda + 1 & 6\lambda - 2 \end{vmatrix} = 0$$

[Expand by R_2]

if,

$$(\lambda - 3) \begin{vmatrix} \lambda - 1 & 5\lambda + 1 \\ 2 & 2(3\lambda + 1) \end{vmatrix} = 0 \quad \text{or if, } 2(\lambda - 3) [(\lambda - 1)(3\lambda - 1) - (5\lambda + 1)] = 0$$

if,

$$6\lambda(\lambda - 3)^2 = 0 \quad \text{or if, } \lambda = 0 \quad \text{or } 3.$$

(a) When $\lambda = 0$, the equations become $-x + y = 0$... (i)

$$-x - 2y + 3z = 0 \quad \dots (ii)$$

$$2x + y - 3z = 0 \quad \dots (iii)$$

Solving (ii) and (iii), we get $\frac{x}{6-3} = \frac{y}{6-3} = \frac{z}{-1+4}$. Hence $x = y = z$.

(b) When $\lambda = 3$, equations becomes identical.

PROBLEMS 2.7

Investigate for consistency of the following equations and if possible find the solutions :

$$4x - 2y + 6z = 8, x + y - 3z = -1, 15x - 3y + 9z = 21.$$

For what values of k the equations $x + y + z = 1, 2x + y + 4z = k, 4x + y + 10z = k^2$ have a solution and solve them completely in each case. (Bhopal, 2008 ; Mumbai, 2008 ; V.T.U., 2006)

Investigate for what values of λ and μ the simultaneous equations

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu,$$

(i) no solution, (ii) a unique solution, (iii) an infinite number of solutions.

(Mumbai, 2007 ; U.P.T.U., 2006 ; Rohtak, 2004)

Investigate for consistency and solve,

$$-3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32. \quad (\text{Bhopal, 2009 ; Kurukshetra, 2005 ; Raipur, 2005})$$

$$2y + z = 3, 2x + 3y + 2z = 5, 3x - 5y + 5z = 2, 3x + 9y - z = 4. \quad (\text{Bhilai, 2005 ; Madras, 2002})$$

$$-6y + 11 = 0, 6x + 20y - 6z + 3 = 0, 6y - 18z + 1 = 0. \quad (\text{Rajasthan, 2005})$$

$$3y + 2z = 1, x + 2y = 4, 10y + 3z = -2, 2x - 3y - z = 5. \quad (\text{U.T.U., 2010 ; Nagarjuna, 2008})$$

5. Find the values of a and b for which the equations

$$x + ay + z = 3, x + 2y + 2z = b, x + 5y + 3z = 9$$

are consistent. When will these equations have a unique solution?

(Kurukshetra, 2005 ; Madras, 2003)

6. Show that if $\lambda \neq -5$, the system of equations

$$3x - y + 4z = 3, x + 2y - 3z = -2, 6x + 5y + \lambda z = -3,$$

have a unique solution. If $\lambda = -5$, show that the equations are consistent. Determine the solutions in each case.

7. Show that the equations

$$3x + 4y + 5z = a, 4x + 5y + 6z = b, 5x + 6y + 7z = c$$

do not have a solution unless $a + c = 2b$.

(Raipur, 2004 ; Nagpur, 2001)

8. Prove that the equations $5x + 3y + 2z = 12, 2x + 4y + 5z = 2, 39x + 43y + 45z = c$ are incompatible unless $c = 74$; and in that case the equations are satisfied by $x = 2 + t, y = 2 - 3t, z = -2 + 2t$, where t is any arbitrary quantity.

9. Find the values of λ for which the equations $(2 - \lambda)x + 2y + 3 = 0, 2x + (4 - \lambda)y + 7 = 0, 2x + 5y + (6 - \lambda)z = 0$ are consistent and find the values of x and y corresponding to each of these values of λ .

10. Show that there are three real values of λ for which the equations $(a - \lambda)x + by + cz = 0, bx + (c - \lambda)y + az = 0, cx + ay + (b - \lambda)z = 0$ are simultaneously true and that the product of these values of λ is

$$D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

11. Determine the values of k for which the following system of equations has non-trivial solutions and find them :

$$(k - 1)x + (4k - 2)y + (k + 3)z = 0, (k - 1)x + (3k + 1)y + 2kz = 0, 2x + (3k + 1)y + 3(k - 1)z = 0.$$

(Mumbai, 2005)

2. Show that the system of equations $2x_1 - 2x_2 + x_3 = \lambda x_1, 2x_1 - 3x_2 + 2x_3 = \lambda x_2, -x_1 + 2x_2 = \lambda x_3$ can possess a non-trivial solution only if $\lambda = 1, \lambda = -3$. Obtain the general solution in each case.

3. Determine the values of λ for which the following set of equations may possess non-trivial solution :

$$3x_1 + x_2 - \lambda x_3 = 0, 4x_1 - 2x_2 - 3x_3 = 0, 2\lambda x_1 + 4x_2 + \lambda x_3 = 0.$$

For each permissible value of λ , determine the general solution.

(Kurukshetra, 2006)

4. Solve completely the system of equations

$$(i) x + y - 2z + 3w = 0; x - 2y + z - w = 0; 4x + y - 5z + 8w = 0; 5x - 7y + 2z - w = 0.$$

$$(ii) 3x + 4y - z - 6w = 0; 2x + 3y + 2z - 3w = 0; 2x + y - 14z - 9w = 0; x + 3y + 13z + 3w = 0. \quad (J.N.T.U., 2002 S)$$

1 (1) LINEAR TRANSFORMATIONS

Let (x, y) be the co-ordinates of a point P referred to set of rectangular axes OX, OY . Then its co-ordinates referred to OX', OY' , obtained by rotating the former axes through an angle θ given by

$$\left. \begin{aligned} x' &= x \cos \theta + y \sin \theta, \\ y' &= -x \sin \theta + y \cos \theta \end{aligned} \right\}$$

A more general transformation than (i) is

$$\left. \begin{aligned} x' &= a_1x + b_1y \\ y' &= a_2x + b_2y \end{aligned} \right\}$$

in matrix notation is $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Such transformations as (i) and (ii), are called *linear transformations* in two dimensions.

Similarly, the relations of the type

$$\left. \begin{aligned} x' &= l_1x + m_1y + n_1z \\ y' &= l_2x + m_2y + n_2z \\ z' &= l_3x + m_3y + n_3z \end{aligned} \right\}$$

linear transformation from (x, y, z) to (x', y', z') in three dimensional problems.

In general, the relation $Y = AX$ where $Y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}, A = \begin{bmatrix} a_1 & b_1 & c_1 & \dots & k_1 \\ a_2 & b_2 & c_2 & \dots & k_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_n & b_n & c_n & \dots & k_n \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$

give linear transformation from n variables $x_1, x_2, \dots, x_n$ to the variables $y_1, y_2, \dots, y_n$ i.e., the transformation of the vector X to the vector Y .

This transformation is called linear because the linear relations $A(X_1 + X_2) = AX_1 + AX_2$ and $A(bX) = bAX$, hold for this transformation.

If the transformation matrix A is singular, the transformation also is said to be singular otherwise non-singular. For a non-singular transformation $Y = AX$, we can also write the inverse transformation $X = A^{-1}Y$. A non-singular transformation is also called a regular transformation.

Cor. If a transformation from (x_1, x_2, x_3) to (y_1, y_2, y_3) is given by $Y = AX$ and another transformation of (y_1, y_2, y_3) to (z_1, z_2, z_3) is given by $Z = BY$, then the transformation from (x_1, x_2, x_3) to (z_1, z_2, z_3) is given by

$$Z = BY = B(AX) = (BA)X.$$

(2) **Orthogonal transformation.** The linear transformation (iv), i.e., $Y = AX$, is said to be **orthogonal** if, it transforms

$$y_1^2 + y_2^2 + \dots + y_n^2 \text{ into } x_1^2 + x_2^2 + \dots + x_n^2$$

The matrix of an orthogonal transformation is called an **orthogonal matrix**.

$$\text{We have } X'X = [x_1 x_2 \dots x_n] \times \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = x_1^2 + x_2^2 + \dots + x_n^2$$

$$\text{and similarly, } Y'Y = y_1^2 + y_2^2 + \dots + y_n^2.$$

$\therefore$ If $Y = AX$ is an orthogonal transformation, then

$$X'X = Y'Y = (AX)'(AX) = X'A'AX \text{ which is possible only if } A'A = I.$$

But $A^{-1}A = I$, therefore, $A' = A^{-1}$ for an orthogonal transformation.

Hence a square matrix A is said to be orthogonal if $AA' = A'A = I$.

Obs. 1. If A is orthogonal, A' and A^{-1} are also orthogonal.

Since A is orthogonal, $A' = A^{-1}$.

$\therefore (A')' = (A^{-1})' = (A')^{-1}$, i.e., $B' = B^{-1}$ where $B = A'$

Hence B (i.e., A') is orthogonal. As $A' = A^{-1}$, A^{-1} is also orthogonal.

Obs. 2. If A is orthogonal, then $|A| = \pm 1$.

$$\text{Since } AA' = A'A = I \quad \therefore |A| |A'| = |I|$$

$$\text{But } |A'| = |A|, \quad \therefore |A| |A| = |1|$$

$$\text{or } |A|^2 = 1 \quad \text{i.e., } |A| = \pm 1.$$

(Mumbai, 2006)

Example 2.38. Show that the transformation

$$y_1 = 2x_1 + x_2 + x_3, y_2 = x_1 + x_2 + 2x_3, y_3 = x_1 - 2x_3$$

is regular. Write down the inverse transformation.

Solution. The given transformation may be written as

$$Y = AX$$

$$\text{where } X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$

$$\text{Now } |A| = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{vmatrix} = -1$$

Thus the matrix A is non-singular and hence the transformation is regular.

$\therefore$ The inverse transformation is given by

$$X = A^{-1}Y$$

$$\text{where } A^{-1} = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix}$$

Thus $x_1 = 2y_1 - 2y_2 - y_3$; $x_2 = -4y_1 + 5y_2 + 3y_3$; $x_3 = y_1 - y_2 - y_3$ is the inverse transformation.

Example 2.39. Prove that the following matrix is orthogonal :

$$\begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix}$$

(Kurukshetra, 2005)

Solution. We have $AA' = \begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix} \times \begin{bmatrix} -2/3 & 2/3 & 1/3 \\ 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \end{bmatrix}$

$$= \begin{bmatrix} 4/9 + 1/9 + 4/9 & -4/9 + 2/9 + 2/9 & -2/9 - 2/9 + 4/9 \\ -4/9 + 2/9 + 2/9 & 4/9 + 4/9 + 1/9 & 2/9 - 4/9 + 2/9 \\ -2/9 - 2/9 + 4/9 & 2/9 - 4/9 + 2/9 & 1/9 + 4/9 + 4/9 \end{bmatrix} = I.$$

Hence the matrix is orthogonal.

Example 2.40. If $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix}$ is orthogonal, find a, b, c and A^{-1} .

(Mumbai, 2006)

Solution. As A is orthogonal, $AA' = I$

$$\therefore \frac{1}{3} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix} \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} 1+4+a^2 & 2+2+ab & 2-4+ac \\ 2+2+ab & 4+1+b^2 & 4-2+bc \\ 2-4+ac & 4-2+bc & 4+4+c^2 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\therefore 5+a^2=9, 5+b^2=9, 8+c^2=9, \text{ i.e., } a^2=4, b^2=4, c^2=1$$

Thus $a=2, b=2, c=1$.

$$\text{Since } A \text{ is orthogonal, } A^{-1} = A' = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & 2 & 1 \end{bmatrix}.$$

$$A^{-1} = A'$$

2.12 (1) VECTORS

Any quantity having n -components is called a *vector of order n* . Therefore, the coefficients in a line equation or the elements in a row or column matrix will form a vector. Thus any n numbers $x_1, x_2, \dots, x_n$ written in a particular order, constitute a vector $\mathbf{x}$.

(2) Linear dependence. The vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are said to be **linearly dependent**, if there exist numbers $\lambda_1, \lambda_2, \dots, \lambda_r$ not all zero, such that

$$\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \dots + \lambda_r \mathbf{x}_r = \mathbf{0}.$$

If no such numbers, other than zero, exist, the vectors are said to be **linearly independent**. If λ_1 transposing $\lambda_1 \mathbf{x}_1$ to the other side and dividing by $-\lambda_1$, we write (i) in the form

$$\mathbf{x}_1 = \mu_2 \mathbf{x}_2 + \mu_3 \mathbf{x}_3 + \dots + \mu_r \mathbf{x}_r.$$

Then the vector $\mathbf{x}_1$ is said to be a linear combination of the vectors $\mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_r$.

Example 2.41. Are the vectors $\mathbf{x}_1 = (1, 3, 4, 2)$, $\mathbf{x}_2 = (3, -5, 2, 2)$ and $\mathbf{x}_3 = (2, -1, 3, 2)$ linearly dependent? If so express one of these as a linear combination of the others.

Solution. The relation $\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \lambda_3 \mathbf{x}_3 = \mathbf{0}$.

$$\lambda_1(1, 3, 4, 2) + \lambda_2(3, -5, 2, 2) + \lambda_3(2, -1, 3, 2) = \mathbf{0}$$

is equivalent to

$$\begin{aligned} \lambda_1 + 3\lambda_2 + 2\lambda_3 &= 0, & 3\lambda_1 - 5\lambda_2 - \lambda_3 &= 0, \\ 4\lambda_1 + 2\lambda_2 + 3\lambda_3 &= 0, & 2\lambda_1 + 2\lambda_2 + 2\lambda_3 &= 0 \end{aligned}$$

As these are satisfied by the values $\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = -2$ which are not zero, the given vectors are linearly dependent. Also we have the relation,

$$\mathbf{x}_1 + \mathbf{x}_2 - 2\mathbf{x}_3 = 0$$

by means of which any of the given vectors can be expressed as a linear combination of the others.

Obs. Applying elementary row operations to the vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, we see that the matrices

$$A = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_1 + \mathbf{x}_2 - 2\mathbf{x}_3 \end{bmatrix}$$

have the same rank. The rank of B being 2, the rank of A is also 2. Moreover $\mathbf{x}_1, \mathbf{x}_2$ are linearly independent and $\mathbf{x}_3$ can be expressed as a linear combination of $\mathbf{x}_1$ and $\mathbf{x}_2$ [$\because \mathbf{x}_3 = \frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2)$]. Similar results will hold for column operations and for any matrix. In general, we have the following results:

If a given matrix has r linearly independent vectors (rows or columns) and the remaining vectors are linear combinations of these r vectors, then rank of the matrix is r . Conversely, if a matrix is of rank r , it contains r linearly independent vectors are remaining vectors (if any) can be expressed as a linear combination of these vectors.

PROBLEMS 2.8

1. Represent each of the transformations

$$x_1 = 3y_1 + 2y_2, y_1 = z_1 + 2z_2 \text{ and } x_2 = -y_1 + 4y_2, y_2 = 3z_1$$

by the use of matrices and find the composite transformation which express x_1, x_2 in terms of z_1, z_2 .

2. If $\xi = x \cos \alpha - y \sin \alpha, \eta = x \sin \alpha + y \cos \alpha$, write the matrix A of transformation and prove that $A^{-1} = A'$. Hence write the inverse transformation.

3. A transformation from the variables x_1, x_2, x_3 to y_1, y_2, y_3 is given by $Y = AX$, and another transformation from y_1, y_2, y_3 to z_1, z_2, z_3 is given by $Z = BY$, where

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -2 \\ -1 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}. \text{ Obtain the transformation from } x_1, x_2, x_3 \text{ to } z_1, z_2, z_3.$$

4. Find the inverse transformation of $y_1 = x_1 + 2x_2 + 5x_3; y_2 = 2x_1 + 4x_2 + 11x_3; y_3 = -x_2 + 2x_3$.

5. Verify that the following matrix is orthogonal:

$$(i) \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix} \quad (\text{Hissar, 2005 S ; P.T.U., 2003})$$

$$(ii) \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

(Kurukshetra, 2005)

6. Find the values of a, b, c if $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal?

(Mumbai, 2005)

7. Prove that $\begin{bmatrix} l & m & n & 0 \\ 0 & 0 & 0 & -1 \\ n & l & -m & 0 \\ -m & n & -l & 0 \end{bmatrix}$ is orthogonal when $l = 2/7, m = 3/7, n = 6/7$.

(Anna, 20)

8. If A and B are orthogonal matrices, prove that AB is also orthogonal.

9. Are the following vectors linearly dependent. If so, find the relation between them:

(Mumbai, 20)

$$(i) (2, 1, 1), (2, 0, -1), (4, 2, 1).$$

$$(ii) (1, 1, 1, 3), (1, 2, 3, 4), (2, 3, 4, 9).$$

$$(iii) \mathbf{x}_1 = (1, 2, 4), \mathbf{x}_2 = (2, -1, 3), \mathbf{x}_3 = (0, 1, 2), \mathbf{x}_4 = (-3, 7, 2).$$

(U.P.T.U., 2003 ; Nagpur, 20)

13 (1) EIGEN VALUES

If A is any square matrix of order n , we can form the matrix $A - \lambda I$, where I is the n th order unit matrix. The determinant of this matrix equated to zero,

is called the *characteristic equation* of A . On expanding the determinant, the characteristic equation takes the form

$$(-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_n = 0,$$

where k 's are expressible in terms of the elements a_{ij} . The roots of this equation are called the *eigenvalues* or *latent roots* or *characteristic roots* of the matrix A .

If $X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$, then the linear transformation $Y = AX$... (i)

carries the column vector X into the column vector Y by means of the square-matrix A . In practice, it is often required to find such vectors which transform into themselves or to a scalar multiple of themselves.

Let X be such a vector which transforms into λX by means of the transformation (i).

Then $\lambda X = AX$ or $AX - \lambda IX = 0$ or $[A - \lambda I]X = 0$... (ii)

This matrix equation represents n homogeneous linear equations

$$\left\{ \begin{array}{l} (a_{11} - \lambda)x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n = 0 \\ \dots\dots\dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + (a_{nn} - \lambda)x_n = 0 \end{array} \right\} \quad \dots(iii)$$

which will have a non-trivial solution only if the coefficient matrix is singular, i.e., if $|A - \lambda I| = 0$.

This is called the characteristic equation of the transformation and is same as the characteristic equation of the matrix A . It has n roots and corresponding to each root, the equation (ii) [or (iii)] will have a non-zero solution.

$X = [x_1, x_2, \dots, x_n]'$, which is known as the *eigen vector* or *latent vector*.

Obs. 1. Corresponding to n distinct eigen values, we get n independent eigen vectors. But when two or more eigen values are equal, it may or may not be possible to get linearly independent eigen vectors corresponding to the repeated root.

Obs. 2. If X_i is a solution for a eigen value λ_i , then it follows from (ii) that cX_i is also a solution, where c is arbitrary constant. Thus the eigen vector corresponding to a eigen value is not unique but may be any one of the vectors cX_i .

Example 2.42. Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. (Bhopal, 200)

Solution. The characteristic equation is $[A - \lambda I] = 0$

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^2 - 7\lambda + 6 = 0$$
$$(\lambda - 6)(\lambda - 1) = 0 \quad \therefore \lambda = 6, 1.$$

Thus the eigen values are 6 and 1.

If x, y be the components of an eigen vector corresponding to the eigen value λ , then

$$[A - \lambda I] X = \begin{bmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Corresponding to $\lambda = 6$, we have $\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

gives only one independent equation - $x + 4y = 0$ ✓

$$\therefore \frac{x}{4} = \frac{y}{1} \text{ giving the eigen vector } (4, 1).$$

Corresponding to $\lambda = 1$, we have $\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$ which gives only one independent equation $x + y = 0$,

$\therefore \frac{x}{1} = \frac{y}{-1}$ giving the eigen vector $(1, -1)$.

Example 2.43. Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.
(Bhopal, 2009; Raipur, 2005)

Solution. The characteristic equation is $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix}$, i.e., $\lambda^3 - 7\lambda^2 + 36 = 0$

Since $\lambda = -2$ satisfies it, we can write this equation as

$(\lambda + 2)(\lambda^2 - 9\lambda + 18) = 0$ or $(\lambda + 2)(\lambda - 3)(\lambda - 6) = 0$.
Thus the eigen values of A are $\lambda = -2, 3, 6$.

If x, y, z be the components of an eigen vector corresponding to the eigen value λ , we have ... (i)

$$[A - \lambda I]X = \begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Putting $\lambda = -2$, we have $3x + y + 3z = 0$, $x + 7y + z = 0$, $3x + y + 3z = 0$.

The first and third equations being the same, we have from the first two

$$\frac{x}{-20} = \frac{y}{0} = \frac{z}{20} \quad \text{or} \quad \frac{x}{-1} = \frac{y}{0} = \frac{z}{1}$$

Hence the eigen vector is $(-1, 0, 1)$. Also every non-zero multiple of this vector is an eigen vector corresponding to $\lambda = -2$.

Similarly, the eigen vectors corresponding to $\lambda = 3$ and $\lambda = 6$ are the arbitrary non-zero multiples of the vectors $(1, -1, 1)$ and $(1, 2, 1)$ which are obtained from (i).

Hence the three eigen vectors may be taken as $(-1, 0, 1)$, $(1, -1, 1)$, $(1, 2, 1)$.

Example 2.44. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ (U.P.T.U., 2005)

Solution. The characteristic equation is

$$[A - \lambda I] = 0, \quad \text{i.e.,} \quad \begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} = 0$$

$$(3 - \lambda)(2 - \lambda)(5 - \lambda) = 0$$

Thus the eigen values of A are 2, 3, 5.

If x, y, z be the components of an eigen vector corresponding to the eigen value λ , we have

$$[A - \lambda I]X = \begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Putting $\lambda = 2$, we have $x + y + 4z = 0$, $6z = 0$, $3z = 0$, i.e., $x + y = 0$ and $z = 0$.

$$\frac{x}{1} = \frac{y}{-1} = \frac{z}{0} = k_1 \text{ (say)}$$

Hence the eigen vector corresponding to $\lambda = 2$ is $k_1(1, -1, 0)$.

Putting $\lambda = 3$, we have $y + 4z = 0$, $-y + 6z = 0$, $2z = 0$, i.e., $y = 0$, $z = 0$.

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{0} = k_2$$

Hence the eigen vector corresponding to $\lambda = 3$ is $k_2 (1, 0, 0)$.
Similarly, the eigen vector corresponding to $\lambda = 5$ is $k_3 (3, 2, 1)$.

2.14 PROPERTIES OF EIGEN VALUES

I. Any square matrix A and its transpose A' have the same eigen values.
We have

$$(A - \lambda I)' = A' - \lambda I' = A' - \lambda I$$

$$| (A - \lambda I)' | = | A' - \lambda I |$$

$$| A - \lambda I | = | A' - \lambda I |$$

$\therefore | A - \lambda I | = 0$ if and only if $| A' - \lambda I | = 0$

i.e., λ is an eigen value of A if and only if it is an eigen value of A' .

II. The eigen values of a triangular matrix are just the diagonal elements of the matrix.

$$\text{Let } A = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{nn} \end{vmatrix} \text{ be a triangular matrix of order } n.$$

Then $| A - \lambda I | = (a_{11} - \lambda)(a_{22} - \lambda) \dots (a_{nn} - \lambda)$.

$\therefore$ Roots of $| A - \lambda I | = 0$ are $\lambda = a_{11}, a_{22}, \dots, a_{nn}$.

Hence the eigen values of A are the diagonal elements of A , i.e., $a_{11}, a_{22}, \dots, a_{nn}$.

Cor. The eigen values of a diagonal matrix are just the diagonal elements of the matrix.

III. The eigen values of an idempotent matrix are either zero or unity.

Let A be an idempotent matrix so that $A^2 = A$. If λ be an eigen value of A , then there exists a non-zero vector X such that

$$AX = \lambda X$$

$$\therefore A(AX) = A(\lambda X), \quad \text{i.e., } A^2X = \lambda(AX)$$

$$AX = \lambda(AX)$$

$$AX = \lambda^2 X$$

From (1) and (2), we get $\lambda^2 X = \lambda X$ or $(\lambda^2 - \lambda) X = 0$

$$\lambda^2 - \lambda = 0 \text{ whence } \lambda = 0 \text{ or } 1.$$

or

Hence the result.

IV. The sum of the eigen values of a matrix is the sum of the elements of the principal diagonal.

[This property will be proved for a matrix of order 3, but the method will be capable of easy extension to matrices of any order.]

Consider the square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$| A - \lambda I | = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$$

so that

$$= -\lambda^3 + \lambda^2(a_{11} + a_{22} + a_{33}) - \dots$$

$$\begin{aligned} \text{If } \lambda_1, \lambda_2, \lambda_3 \text{ be the eigen values of } A, \text{ then } | A - \lambda I | &= (-1)^3 (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) \\ &= -\lambda^3 + \lambda^2(\lambda_1 + \lambda_2 + \lambda_3) - \dots \end{aligned}$$

Equating the right hand sides of (ii) and (iii) and comparing coefficients of λ^2 , we get

$$\lambda_1 + \lambda_2 + \lambda_3 = a_{11} + a_{22} + a_{33}. \text{ Hence the result.}$$

V. The product of the eigen values of a matrix A is equal to its determinant.

Putting $\lambda = 0$ in (iii), we get the result.

VI. If λ is an eigen value of a matrix A , then $1/\lambda$ is the eigen value of A^{-1} .

If X be the eigen vector corresponding to λ , then $AX = \lambda X$

Premultiplying both sides by A^{-1} , we get $A^{-1}AX = A^{-1}\lambda X$, i.e., $A^{-1}X = (1/\lambda)X$, i.e., $X = \lambda(A^{-1}X)$, i.e., X is an eigen value of the inverse matrix A^{-1} .

$IX = \lambda A^{-1}X$ or $X = \lambda(A^{-1}X)$, i.e., X is an eigen value of A^{-1} . [Property VI]. Since A is an orthogonal matrix, A^{-1} is same as its transpose A' .

VII. If λ is an eigen value of an orthogonal matrix, then $1/\lambda$ is also its eigen value. This being of the same form as (i), shows that $1/\lambda$ is an eigen value of A^{-1} . [Property VI]. Since A is an orthogonal matrix, A^{-1} is same as its transpose A' .

$\therefore 1/\lambda$ is an eigen value of A' .

But the matrices A and A' have the same eigen values, since the determinants $|A - \lambda J|$ and $|A' - \lambda J|$ are the same.

Hence $1/\lambda$ is also an eigen value of A .

VIII. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigen values of a matrix A , then A^m has the eigen values $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ (m being a positive integer).

Let A_i be the eigen value of A and X_i the corresponding eigen vector. Then $AX_i = \lambda_i X_i$ which is of the same form as (i).

We have

$$A^2 X_i = A(AX_i) = A(\lambda_i X_i) = \lambda_i (AX_i) = \lambda_i (\lambda_i X_i) = \lambda_i^2 X_i$$

Similarly,

$$A^3 X_i = \lambda_i^3 X_i$$

Hence λ_i^m is an eigen value of A^m .

The corresponding eigen vector is the same X_i .

2.15 CAYLEY-HAMILTON THEOREM*

Every square matrix satisfies its own characteristic equation; i.e., if the characteristic equation for the n th order square matrix A is

$$|A - \lambda J| = (-1)^n \lambda^n + k_1 \lambda^{n-1} + \dots + k_n = 0$$

then

$$(-1)^n A^n + k_1 A^{n-1} + \dots + k_n I = 0.$$

Let the adjoint of the matrix $A - \lambda J$ be P . Clearly, the elements of P will be polynomials of the $(n-1)$ th degree in λ , for the cofactors of the elements in $|A - \lambda J|$ will be such polynomials.

$\therefore P$ can be split up into a number of matrices, containing terms with the same powers of λ , such that

$$P = P_1 \lambda^{n-1} + P_2 \lambda^{n-2} + \dots + P_{n-1} \lambda + P_n \quad \dots(ii)$$

where $P_1, P_2, \dots, P_n$ are all the square matrices of order n whose elements are functions of the elements of A .

Since the product of a matrix by its adjoint = determinant of the matrix $\times$ unit matrix.

$$[A - \lambda J]P = |A - \lambda J| \times I$$

$\therefore$ by (i) and (ii),

$$[A - \lambda J] [P_1 \lambda^{n-1} + P_2 \lambda^{n-2} + \dots + P_{n-1} \lambda + P_n]$$

$$= [(-1)^n \lambda^n + k_1 \lambda^{n-1} + \dots + k_{n-1} \lambda + k_n] I.$$

Equating the coefficients of various powers of λ , we get

$$-P_1 = (-1)^n I$$

$$AP_1 - P_2 = k_1 I,$$

$$AP_2 - P_3 = k_2 I,$$

$$AP_{n-1} - P_n = k_{n-1} I,$$

$$AP_n = k_n I.$$

Now pre-multiplying the equations by $A^n, A^{n-1}, \dots, A, I$ respectively and adding, we get

$$(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = 0,$$

terms on the left cancel in pairs. This proves the theorem.

Cor. Another method of finding the inverse. Multiplying (iii) by A^{-1} , we get

$$(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I + k_n A^{-1} = 0$$

$$A^{-1} = -\frac{1}{k_n} [(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I].$$

note on p.17. William Rowan Hamilton (1805-1865) an Irish mathematician who is known for his work i

This result gives the inverse of A in terms of $n-1$ powers of A and is considered as a practical method for the computation of the inverse of the large matrices. As a by-product of the computation, the characteristic equation and the determinant of the matrix are also obtained.

Example 2.45. Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find its inverse.

Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A . (Bhopal, 2009)

Solution. The characteristic equation of A is

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^2 - 4\lambda - 5 = 0 \quad \dots(i)$$

By Cayley-Hamilton theorem, A must satisfy its characteristic equation (i), so that

$$A^2 - 4A - 5I = 0 \quad \dots(ii)$$

Now

$$\begin{aligned} A^2 - 4A - 5I &= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \end{aligned}$$

This verifies the theorem.

Multiplying (ii) by A^{-1} , we get $A - 4I - 5A^{-1} = 0$

$$\text{or} \quad A^{-1} = \frac{1}{5}(A - 4I) = \frac{1}{5} \left\{ \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

Now dividing the polynomial $\lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10I$ by the polynomial $\lambda^2 - 4\lambda - 5$, we obtain

$$\begin{aligned} \lambda^5 - 4\lambda^4 - 7\lambda^3 - \lambda - 10I &= (\lambda^2 - 4\lambda - 5)(\lambda^3 + 2\lambda + 3) + \lambda + 5 \\ &= \lambda + 5 \end{aligned} \quad [\text{By (i)}]$$

Hence $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = A + 5$, which is a linear polynomial in A .

Example 2.46. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ and hence find its inverse.

$$\text{Solution. The characteristic equation is } \begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 3-\lambda & -3 \\ -2 & -4 & -4-\lambda \end{vmatrix} = 0, \quad \text{i.e., } \lambda^3 - 20\lambda + 8 = 0.$$

By Cayley-Hamilton theorem, $A^3 - 20A + 8I = 0$, whence $A^{-1} = \frac{5}{2}I - \frac{1}{8}A^2$,

$$= \frac{5}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{8} \begin{bmatrix} -4 & -8 & -12 \\ 10 & 22 & 6 \\ 2 & 2 & 22 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 3/2 \\ -5/4 & -1/4 & -3/4 \\ -1/4 & -1/4 & -1/4 \end{bmatrix} \quad [\text{cf. Ex.}]$$

Example 2.47. Find the characteristic equation of the matrix, $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} (U.T.U.,)

Also find the matrix represented by

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I. \quad \left(\text{Anna, 2009 ; Rajasthan, 2005 ; U.P.T.U.,} \right)$$

Solution. The characteristic equation of the matrix A is

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

ing to Cayley-Hamilton theorem, we have $A^3 - 5A^2 + 7A - 3I = 0$
 multiplying (i) by A^{-1} , we get

$$A^2 - 5A + 7I - 3A^{-1} = 0 \quad \text{or} \quad A^{-1} = \frac{1}{3} [A^2 - 5A + 7I]$$

But

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4+0+1 & 2+1+1 & 2+0+2 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 2+0+2 & 1+1+2 & 1+0+4 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$\therefore A^2 - 5A + 7I = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 5 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

Hence from (ii), $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

Now

$$\begin{aligned} A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I \\ = A^5 (A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I \\ = A^2 + A + I \end{aligned}$$

$$[\because A^3 - 5A^2 + 7A - 3I = 0]$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

PROBLEMS 2.9

- Find the sum and product of the eigen values of $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

(a) $\begin{bmatrix} 4 & 3 \\ 2 & 9 \end{bmatrix}$

- Find the eigen values and eigen vectors of the matrices:

(W.B.T.U., 2005)

(b) $\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$

- Find the latent roots and the latent vectors of the matrices:

(a) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

be an eigen value of a non-singular matrix A , show that $|A|/\lambda$ is an eigen value of the matrix $\text{adj } A$.

the eigen values of $\text{adj } A$ and of $A^2 - 2A + I$, where $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$

en values of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are = 1 each. Find the eigen values of A^{-1} .

if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the latent roots of a matrix A , then A^2 has the latent roots $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$.

(Madras, 2006)

(Bhopal, 2002 S)

(Bhopal, 2008; Nagarjuna, 2008; S.V.T.U., 2008; J.N.T.U., 2006)

(J.N.T.U., 2005; Kurukshetra, 2005)

(Mumbai, 2006; B.P.T.U., 2006; U.P.T.U., 2006)

(Madras, 2006)

(U.P.T.U., 2001)

(Mumbai, 2006)

(P.T.U., 2005)

8. For a symmetrical square matrix, show that the eigen vectors corresponding to two unequal eigen values are orthogonal.

9. Using Cayley-Hamilton theorem, find the inverse of

$$(i) \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

(Osmania, 2000 S)

$$(iii) \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ 2 & -4 & -4 \end{bmatrix}$$

(Bhopal, 2002 S)

$$(iv) \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

(U.P.T.U., 2006)

10. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. Show that the equation is satisfied by A and hence obtain the inverse of the given matrix.

(Bhopal, 2008 ; Anna, 2005 ; Kerala, 2005)

11. Verify Cayley-Hamilton theorem for the matrix A and find its inverse.

$$(i) \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

(Anna, 2009 ; S.V.T.U., 2008 ; Madras, 2006)

$$(ii) \begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$$

(Coimbatore, 2001)

$$(iii) \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix}$$

(P.T.U., 2006)

12. Using Cayley-Hamilton theorem, find A^8 , if $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.

(Anna, 2003)

13. If $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, find A^4 .

(Madras, 2000)

14. Using Cayley-Hamilton theorem, find A^{-2} , where $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

(Bhopal, 2000)

15. If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$, evaluate A^{-1} , A^{-2} and A^{-3} .

16. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, show that $A^n = A^{n-2} + A^2 - I$. Hence find A^{50} .

(Mumbai, 2000)

2.16 (1) REDUCTION TO DIAGONAL FORM

If a square matrix A of order n has n linearly independent eigen vectors, then a matrix P can be found such that $P^{-1}AP$ is a diagonal matrix.

[This result will be proved for a square matrix of order 3 but the method will be capable of easy extension to any order.]

Let A be a square matrix of order 3. Let $\lambda_1, \lambda_2, \lambda_3$ be its eigen values and

$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be the corresponding eigen vectors.

Denoting the square matrix $[X_1 X_2 X_3] = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$ by P , we have

$$AP = A[X_1 X_2 X_3] = [AX_1, AX_2, AX_3] = [\lambda_1 X_1, \lambda_2 X_2, \lambda_3 X_3]$$

$$= \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \times \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = PD, \text{ where } D \text{ is the diagonal matrix.}$$

$\therefore P^{-1}AP = P^{-1}PD = D$, which proves the theorem.

Obs. 1. The matrix P which diagonalises A is called the **modal matrix** of A and the resulting diagonal matrix D is known as the **spectral matrix** of A .

2. The diagonal matrix has the eigen values of A as its diagonal elements.

3. The matrix P , which diagonalise A , constitutes the eigen vectors of A .

(2) Similarity of matrices. A square matrix $\hat{A}$ of order n is called **similar** to a square matrix A of order n if

$$\hat{A} = P^{-1}AP \text{ for some non-singular } n \times n \text{ matrix } P.$$

This transformation of a matrix A by a non-singular matrix P to $\hat{A}$ is called a **similarity transformation**.

Obs. If the matrix $\hat{A}$ is similar to the matrix A , then $\hat{A}$ has the same eigen values as A .

If $\mathbf{x}$ is an eigen vector of A , then $\mathbf{y} = P^{-1}\mathbf{x}$ is an eigen vector of $\hat{A}$ corresponding to the same eigen value.

(3) Powers of a matrix. Diagonalisation of a matrix is quite useful for obtaining powers of a matrix.

Let A be the square matrix. Then a non-singular matrix P can be found such that

$$D = P^{-1}AP$$

$$D^2 = (P^{-1}AP)(P^{-1}AP) = P^{-1}A^2P$$

$$D^3 = P^{-1}A^3P \text{ and in general, } D^n = P^{-1}A^nP$$

Similarly,

To obtain A^n , premultiply (i) by P and post-multiply by P^{-1} .

Then $PD^nP^{-1} = PP^{-1}A^nPP^{-1} = A^n$ which gives A^n .

$$[\because PP^{-1} = I] \quad \dots(i)$$

Thus,

$$A^n = PD^nP^{-1} \text{ where, } D^n = \begin{bmatrix} \lambda_1^n & 0 & 0 \\ 0 & \lambda_2^n & 0 \\ 0 & 0 & \lambda_3^n \end{bmatrix}$$

Working procedure :

1. Find the eigen values of the square matrix A .
2. Find the corresponding eigen vectors and write the modal matrix P .
3. Find the diagonal matrix D from $D = P^{-1}AP$
4. Obtain A^n from $A^n = PD^nP^{-1}$.

Example 2.48. Reduce the matrix $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ to the diagonal form.

(V.T.U., 2011 ; U.T.U., 2010 ; Bhopal, 2009 ; U.P.T.U., 2000)

Solution. The characteristic equation of A is

$$\begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^3 - \lambda^2 - 5\lambda + 5 = 0.$$

Solving, we get $\lambda_1 = 1, \lambda_2 = \sqrt{5}, \lambda_3 = -\sqrt{5}$ as the eigen values of A .

When $\lambda = 1$, the corresponding eigen vector is given by

$$-2x + 2y - 2z = 0, x + y + z = 0, -x - y - z = 0.$$

Solving the first two equations, we get $\frac{x}{2} = \frac{y}{0} = \frac{z}{-2}$ giving the eigen vector $(1, 0, -1)$

When $\lambda = \sqrt{5}$, the corresponding eigen vector is given by

$$(-1 - \sqrt{5})x + 2y - 2z = 0, x + (2 - \sqrt{5})y + z = 0, -x - y - \sqrt{5}z = 0.$$

Solving 2nd and 3rd equations, we get

$$\frac{x}{6-2\sqrt{5}} = \frac{y}{-1+\sqrt{5}} = \frac{z}{1-\sqrt{5}} \quad \text{or} \quad \frac{x}{\sqrt{5}-1} = \frac{y}{1} = \frac{z}{-1}$$

ng the eigen vector $(\sqrt{5}-1, 1, -1)$.

Similarly the eigen vector corresponding to $\lambda = -\sqrt{5}$, is $(\sqrt{5}+1, -1, 1)$.

Writing the three eigen vectors as the three columns, we get the transformation (*modal*) matrix as

$$P = \begin{bmatrix} 1 & \sqrt{5}-1 & \sqrt{5}+1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}.$$

Hence the diagonal matrix is

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & -\sqrt{5} \end{bmatrix}.$$

Example 2.49. Find the matrix P which transforms the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ to the diagonal form.

calculate A^4 .

Solution. The eigen values of A (found in Ex. 2.43) are $-2, 3, 6$ and the eigen vectors are $(-1, 0, 1)$, $(1, 2, 1)$. Writing these eigen vectors as the three columns, the required transformation matrix (*modal*) is

$$P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

find P^{-1} ,

$$|P| = \begin{vmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad (\text{say})$$

$$A_1 = -3, B_1 = 2, C_1 = 1, A_2 = 0, B_2 = -2, C_2 = 2, A_3 = 3, B_3 = 2, C_3 = 1$$

so

$$|P| = a_1 A_1 + b_1 B_1 + c_1 C_1 = 6$$

$$P^{-1} = \frac{1}{|P|} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

is

$$D = P^{-1}AP = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$D^4 = \begin{bmatrix} (-2)^4 & 0 & 0 \\ 0 & 3^4 & 0 \\ 0 & 0 & 6^4 \end{bmatrix} = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 1296 \end{bmatrix}$$

e

$$A^4 = PD^4P^{-1} = \frac{1}{6} \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 16 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 1296 \end{bmatrix} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -8 & 0 & 8 \\ 27 & -27 & 27 \\ 216 & 512 & 216 \end{bmatrix} = \begin{bmatrix} 251 & 485 & 235 \\ 485 & 1051 & 485 \\ 235 & 485 & 251 \end{bmatrix}$$

Example 2.50. Find e^A and 4^A if $A = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix}$.

Solution. The characteristic equation of A is

$$\begin{vmatrix} 3/2 - \lambda & 1/2 \\ 1/2 & 3/2 - \lambda \end{vmatrix} = 0, \text{ i.e., } (3/2 - \lambda)^2 - 1/4 = 0.$$

$$\lambda^2 - 3\lambda + 2 = 0 \text{ whence } \lambda = 1, 2.$$

When $\lambda = 1$, $[A - \lambda I] X = 0$, gives

$$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore x_1 + x_2 = 0$. If $x_2 = -1$, $x_1 = 1$, i.e., the eigen vector is $[1, -1]'$.

When $\lambda = 2$, $[A - \lambda I] X = 0$, gives

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$-x_1 + x_2 = 0, \text{ i.e., } x_1 = x_2$$

If $x_2 = 1$, $x_1 = 1$, i.e., the eigen vector is $[1, 1]'$

Now $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

$\therefore$

$$P^{-1} = \frac{\text{adj } P}{|P|} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

If

$$f(A) = e^A, f(D) = e^D = \begin{bmatrix} e^1 & 0 \\ 0 & e^2 \end{bmatrix}$$

$\therefore$

$$e^A = P f(D) P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} e^1 & 0 \\ 0 & e^2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} e & e^2 \\ -e & e^2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} e + e^2 & -e + e^2 \\ -e + e^2 & e + e^2 \end{bmatrix}$$

Replacing e by 4, we get

$$4^A = \frac{1}{2} \begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}.$$

[By $2R_1, 2R_2$]

[By $R_2 - R_1$]

[By $2R_1, 2R_2$]

[By $R_2 - R_1$]

2.17 REDUCTION OF QUADRATIC FORM TO CANONICAL FORM

A homogeneous expression of the second degree in any number of variables is called a *quadratic form*.

For instance, if $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $X' = [x \ y \ z]$, then

$$X'AX = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$$

is a *quadratic form*.

Let $\lambda_1, \lambda_2, \lambda_3$ be the eigen values of the matrix A and

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}, X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$$

its corresponding eigen vectors in the normalized form (i.e., each element is divided by square root of sum of squares of all the three elements in the eigen vector).

Then by § 2.16(1), $P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ where $P = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$

Hence the quadratic form (i) is reduced to a canonical form (or sum of squares form or Principal axes form).

$$\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$$

where P is the matrix of transformation which is an orthogonal matrix.

Note. Congruent (or orthogonal) transformation. The diagonal matrix D and the matrix A are called congruent matrices and the above method of reduction is called congruent (or orthogonal) transformation.

Remember that the matrix A corresponding to the quadratic form

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$$

$$\begin{bmatrix} \text{coeff. of } x^2 & \frac{1}{2} \text{ coeff. of } yz & \frac{1}{2} \text{ coeff. of } zx \\ \frac{1}{2} \text{ coeff. of } yz & \text{coeff. of } y^2 & \frac{1}{2} \text{ coeff. of } xy \\ \frac{1}{2} \text{ coeff. of } zx & \frac{1}{2} \text{ coeff. of } xy & \text{coeff. of } z^2 \end{bmatrix}, \text{ i.e., } \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}.$$

Example 2.51. Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form and specify the matrix of transformation. (Bhopal, 2009; Kurukshetra, 2006)

Solution. The matrix of the given quadratic form is $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Its characteristic equation is $|A - \lambda I| = 0$, i.e., $\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$

which gives $\lambda = 2, 3, 6$ as its eigen values. Hence the given quadratic form reduces to the canonical form

$$\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2, \text{ i.e., } 2x^2 + 3y^2 + 6z^2.$$

To find the matrix of transformation

From $[A - \lambda I] X = 0$, we obtain the equations

$$(3 - \lambda)x - y + z = 0; -x + (5 - \lambda)y - z = 0; x - y + (3 - \lambda)z = 0.$$

Now corresponding to $\lambda = 2$, we get $x - y + z = 0$, $-x + 3y - z = 0$, and $x - y + z = 0$,

we

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$$

$\therefore$ The eigen vector is $X_1 (1, 0, -1)$ and its normalised form is $(1/\sqrt{2}, 0, -1/\sqrt{2})$.

Similarly, corresponding to $\lambda = 3$, the eigen vector is $X_2 (1, 1, 1)$ and its normalised form is $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$.

Finally, corresponding to $\lambda = 6$, the eigen vector is $X_3 (1, -2, 1)$ and its normalised form is $(1/\sqrt{6}, -2/\sqrt{6}, 1/\sqrt{6})$.

Hence the matrix of transformation is $P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}.$

2.18 NATURE OF A QUADRATIC FORM

Let $Q = X'AX$ be a quadratic form in n variables $x_1, x_2, \dots, x_n$.

Index. The number of positive terms in its canonical form is called the index of the quadratic form.

Signature (S) of the quadratic form is the difference of positive and negative terms in the canonical form. If the rank of the matrix A is r and the signature of the quadratic form Q is s , then the quadratic form is said to be

- (i) positive definite if $r = n$ and $s = n$
- (ii) negative definite if $r = n$ and $s = 0$
- (iii) positive semidefinite if $r < n$ and $s = r$
- (iv) negative semidefinite if $r < n$ and $s = 0$
- (v) indefinite in all other cases.

In other words a real quadratic form $X'AX$ in n variables is said to be

- (i) **positive definite** if all the eigen values of $A > 0$.
- (ii) **negative definite** if all the eigen values of $A < 0$.
- (iii) **positive semidefinite** if all the eigen values of $A \geq 0$ and at least one eigen value $= 0$.
- (iv) **negative semidefinite** if all the eigen values of $A \leq 0$ and at least one eigen value $= 0$.
- (v) **indefinite** if some of the eigen values of A are positive and others negative.

Example 2.52. Reduce the quadratic form $2x_1x_2 + 2x_1x_3 - 2x_2x_3$ to a canonical form by an orthogonal reduction and discuss its nature. (Madras, 2006)

Also find the modal matrix.

Solution. (i) The matrix of the given quadratic form is $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

Its characteristic equation is $[A - \lambda I] = 0$, i.e., $\begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{bmatrix} = 0$

which gives $\lambda^3 - 3\lambda + 2 = 0$

Solving, we get $\lambda = 1, 1, -2$ as the eigen values. Hence the given quadratic form reduces to the canonical form

$$\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2 = 0, \text{ i.e., } x^2 + y^2 - 2z^2 = 0$$

(ii) Since some of the eigen values of A are positive and others are negative, the given quadratic form is **Indefinite**.

(iii) To find the matrix of transformation

From $[A - \lambda I] X = 0$, we get the equations

$$-\lambda x + y + z = 0, x - \lambda y + z = 0, x - y - \lambda z = 0$$

When $\lambda = -2$, we get $2x + y + z = 0, x + 2y - z = 0, x - y + 2z = 0$.

Solving first and second equations, we get

$$\frac{x}{-1} = \frac{y}{1} = \frac{z}{1}$$

$\therefore$ The corresponding eigen vector $X_1 = (-1, 1, 1)$ and its normalised form is $(-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$

When $\lambda = 1$, we get $-x + y + z = 0, x - y - z = 0, x - y - z = 0$.

These equations are same. Take $y = 0$ so that $x = z$.

$\therefore$ The corresponding eigen vector $X_2 = (1, 0, 1)$ and its normalised form is $(1/\sqrt{2}, 0, 1/\sqrt{2})$

To find the eigen vector $X_3 = (l, m, n)$ (say)

Since X_3 is orthogonal to X_1 , $\therefore -l + m + n = 0$

Since X_3 is orthogonal to X_2 , $\therefore l + n = 0$

These equations give $\frac{l}{1} = \frac{m}{2} = \frac{n}{-1}$.

The eigen vector $X_3 = (1, 2, -1)$ and normalised form is $(1/\sqrt{6}, 2/\sqrt{6}, -1/\sqrt{6})$.

Hence the modal matrix is

$$P = \begin{bmatrix} -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

PROBLEMS 2.10

1. If $A = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $P = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $P^{-1}AP$ is a diagonal matrix.

2. Show that the linear transformation

$$H = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ where } \theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}, \text{ changes the matrix}$$

$$C = \begin{bmatrix} a & h \\ h & b \end{bmatrix} \text{ to the diagonal form } D = HCH'.$$

3. Reduce the matrix $A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$ to the diagonal form.

(B.P.T.U., 2005)

4. If $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, find A^n and A^4 .

(Mumbai, 2006)

5. If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, calculate A^4 .

(Coimbatore, 2001)

6. If $A = \begin{bmatrix} -1 & 4 \\ 2 & 1 \end{bmatrix}$, then prove that $3 \tan A = A \tan 3$.

(Mumbai, 2006)

7. Find the eigen vectors of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ and hence reduce $6x^2 + 3y^2 + 3z^2 - 2yz + 4zx - 4xy$ to a 'sum of squares'. Also write the nature of the matrix.

(Calicut, 2004)

8. Reduce the quadratic form $2xy + 2yz + 2zx$ into canonical form.

(Anna, 2009 ; Kurukshetra, 2006 ; Mumbai, 2006)

9. (a) Find the eigen values, eigen vectors and the modal of matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$.

(b) Reduce the quadratic form $x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$ to a canonical form.

(Anna, 2006)

10. Reduce the following quadratic forms into a 'sum of squares' by an orthogonal transformation and give the matrix of transformation. Also state the nature of each of these.

(i) $3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3$.

(ii) $8x^2 + 7y^2 + 3z^2 - 12xy - 8yz + 4zx$

(Anna, 2006)

1. Find the index and signature of the quadratic form $x_1^2 + 2x_2^2 - 3x_3^2$.

(Madras, 2006)

2. Find the nature of the quadratic form $x^2 + 5y^2 + z^2 + 2xy + 2yz + 6zx$.

(Bhopal, 2006)

3. Show that the form $5x_1^2 + 26x_2^2 + 10x_3^2 + 4x_2x_3 + 14x_3x_1 + 6x_1x_2$ is a positive semi-definite and find a non-zero values of x_1, x_2, x_3 which make the form zero.

(P.T.U., 2006)

9 COMPLEX MATRICES

So far, we have considered matrices whose elements were real numbers. The elements of a matrix, however, be complex numbers also.

(1) Conjugate of a matrix. If the elements of a matrix $A = [a_{rs}]$ are complex numbers $\alpha_{rs} + i\beta_{rs}$, α_{rs} real, then the matrix

$\bar{A} = [\bar{a}_{rs}] = [\alpha_{rs} - i\beta_{rs}]$ is called the **conjugate matrix** of A .

The transpose of a conjugate of a matrix A is denoted by A^* or A^H , i.e., $(\bar{A})' = A^*$.

(2) **Hermitian matrix.** A square matrix A such that $A' = \bar{A}$ is said to be a **Hermitian matrix**. The elements of the leading diagonal of a Hermitian matrix are evidently real, while every other element is the complex conjugate of the element in the transposed position. For instance $A = \begin{bmatrix} 2 & 3+4i \\ 3-4i & -5 \end{bmatrix}$ is a Hermitian

matrix, since $A' = \begin{bmatrix} 2 & 3-4i \\ 3+4i & -5 \end{bmatrix} = \bar{A}$

(3) **Skew-Hermitian matrix.** A square matrix A such that $A' = -\bar{A}$ is said to be a **skew-Hermitian matrix**. This implies that the leading diagonal elements of a skew-Hermitian matrix are either all zeros or all purely imaginary.

Obs. A Hermitian matrix is a generalisation of a real symmetric matrix as every real symmetric matrix is Hermitian. Similarly, a skew-Hermitian matrix is a generalisation of a real skew-symmetric matrix.

Properties

I. Any square matrix A can be written as the sum of a Hermitian and skew-Hermitian matrices.

(Mumbai, 2007)

Take

$$B = \frac{1}{2}(A + \bar{A}') \text{ and } C = \frac{1}{2}(A - \bar{A}')$$

Then

$$B' = \frac{1}{2}(A + \bar{A}') = \frac{1}{2}(A' + \bar{A})$$

$$\bar{B} = \frac{1}{2}\overline{(A + \bar{A}')} = \frac{1}{2}(\bar{A} + A') = B'$$

and

i.e., B is a Hermitian matrix.

Again,

$$C' = \frac{1}{2}(A - \bar{A}') = \frac{1}{2}(A' - \bar{A})$$

and

$$\bar{C} = \frac{1}{2}\overline{(A - \bar{A}')} = \frac{1}{2}(\bar{A} - A') = -C'$$

$\therefore C' = -C$, i.e., C is a skew-Hermitian matrix.

Thus,

$$A = \frac{1}{2}(A + \bar{A}') + \frac{1}{2}(A - \bar{A}') = B + C$$

Hence the result.

II. If A is a Hermitian matrix, then (iA) is a skew-Hermitian matrix.

(Mumbai, 2007)

We have

$$\begin{aligned} (\overline{iA})' &= (\bar{i} \bar{A}') = (-i \bar{A}') = -i \bar{A}' \\ &= -iA \end{aligned}$$

$[\because \bar{A}' = A]$

Thus (iA) is a skew-Hermitian matrix.

Similarly if A is a skew-Hermitian matrix then (iA) is a Hermitian matrix.

III. The eigen values of a Hermitian matrix are real. (see Fig. 2.1)

Let λ be the eigen value and X the corresponding eigen vector of a Hermitian matrix A , so that

$$AX = \lambda X$$

$$\bar{X}'AX = \bar{X}'\lambda X = \lambda \bar{X}'X \quad \text{or} \quad \lambda = \bar{X}'AX / \bar{X}'X$$

Since $\bar{X}'X = \bar{x}_1x_1 + \bar{x}_2x_2 + \dots + \bar{x}_nx_n = |x_1|^2 + |x_2|^2 + \dots + |x_n|^2$ is real and non-zero. Also $\bar{X}'AX$ is a Hermitian scalar and is always real.

$\therefore \lambda$, the eigen value of a Hermitian matrix is real.

IV. The eigen values of a skew-Hermitian matrix are purely imaginary or zero.

after the French mathematician Charles Hermite (1822–1901), known for his contributions to algebra and number theory.

Let λ be the eigen value and X the corresponding eigen vector of a skew-Hermitian matrix B so that $BX = \lambda X$.

$$\therefore \bar{X}'BX = \bar{X}'\lambda X = \lambda \bar{X}'X \quad \text{or} \quad \lambda = \bar{X}'BX / \bar{X}'X$$

Since $\bar{X}'X$ is real and non-zero. Also $\bar{X}BX$ is a skew-Hermitian form which is purely imaginary or zero.

$\therefore \lambda$, the eigen value of a skew-Hermitian matrix is purely imaginary or zero.

4. Unitary matrix. A square matrix U such that $\bar{U}' = U^{-1}$ is called a **unitary matrix**. For a unitary matrix, $U, U^{-1}, U^* = U^*, U = I$.

This is a generalisation of the orthogonal matrix in the complex field.

Properties

I. Inverse of a unitary matrix is unitary

If U is a unitary matrix, then

$$\bar{U}' = U^{-1}$$

or

$$U' = \overline{U^{-1}}$$

$\therefore$

$$[(U^{-1})^{-1}]' = \overline{U^{-1}}$$

Writing $U^{-1} = V$, we have

$$[V^{-1}]' = \bar{V} \quad \text{or} \quad V^{-1} = \bar{V}'$$

Thus $V (= U^{-1})$ is also unitary.

Cor. Inverse of an orthogonal matrix is orthogonal.

II. Transpose of a unitary matrix is unitary

If U is a unitary matrix,

$$\bar{U}' = U^{-1}$$

or

$$(\bar{U}') = U^{-1}$$

or

$$[(\bar{U}')] = [U^{-1}]' = [U']^{-1}$$

Writing $U' = V$, we have $\bar{V}' = V^{-1}$

Thus V (i.e., U') is also unitary.

Cor. Transpose of an orthogonal matrix is orthogonal.

III. Product of two unitary matrices is a unitary matrix.

If U and V are unitary matrices then

$$U' = \bar{U}^{-1}, V' = \bar{V}^{-1}$$

Now,

$$\begin{aligned} (\overline{UV})^{-1} &= (\overline{UV})^{-1} = \bar{V}^{-1}\bar{U}^{-1} \\ &= V'U' \\ &= (UV)' \end{aligned}$$

[$\because U, V$ are

Thus, UV is a unitary matrix.

Cor. Product of two orthogonal matrices is an orthogonal matrix.

IV. The eigen value of a unitary matrix has absolute value 1.

(U.T.)

If U is a unitary matrix then $UX = \lambda X$

Taking conjugate transpose of (1),

$$(\overline{UX})' = (\overline{UX})' = \bar{X}'\bar{U}' = \bar{X}'\bar{U}^{-1}$$

Also

$$(\overline{UX})' = (\overline{\lambda X})' = \bar{\lambda}\bar{X}'$$

$$\bar{X}'U^{-1} = \bar{\lambda}\bar{X}'$$

Post-multiplying (2) by (1), we get

$$(\bar{X}'U^{-1})(UX) = (\bar{\lambda}\bar{X}')(\lambda X)$$

$$\bar{X}'(U^{-1}U)X = (\bar{\lambda}\lambda)(\bar{X}'X)$$

$$\bar{X}'X = (\lambda\lambda')\bar{X}'X$$

Thus

$$\lambda\lambda' = |\lambda|^2 = 1.$$

Hence the result.

Cor. The eigen value of an orthogonal matrix has absolute value 1.

Example 2.53. If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, show that AA^* is a Hermitian matrix, where A^* is the conjugate transpose of A .

(J.N.T.U., 2005; U.P.T.U., 2003)

Solution. We have $A' = \begin{bmatrix} 2+i & -5 \\ 3 & i \\ -1+3i & 4-2i \end{bmatrix}$

and

$$A^* = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix}$$

$$\therefore AA^* = \begin{bmatrix} 2-i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix} \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix}$$

$$= \begin{bmatrix} 4-i^2+9+1-9i^2 & -10-5i-3i-10+10i \\ -10+5i+3i-10-10i & 25-i^2+16-4i^2 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & -20+2i \\ -20-2i & 46 \end{bmatrix}, \text{ which is a Hermitian matrix.}$$

Example 2.54. Prove that the matrix $A = \begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(-1+i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix}$ is unitary and find A^{-1} .

(Mumbai, 2006)

Solution. Conjugate of A , i.e., $\bar{A} = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(-1-i) \\ \frac{1}{2}(1-i) & \frac{1}{2}(1+i) \end{bmatrix}$

$$\therefore \text{Transpose of } \bar{A}, \text{ i.e., } A^\theta = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ \frac{1}{2}(-1-i) & \frac{1}{2}(1+i) \end{bmatrix}$$

Now

$$\begin{aligned} A^\theta \cdot A &= \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ \frac{1}{2}(-1-i) & \frac{1}{2}(1+i) \end{bmatrix} \begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(-1+i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{4}(1+1) + \frac{1}{4}(1+1) & -\frac{1}{4}(1-i)^2 + \frac{1}{4}(1-i)^2 \\ -\frac{1}{4}(1+i)^2 + \frac{1}{4}(1+i)^2 & \frac{1}{4}(1+1) + \frac{1}{4}(1+1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \end{aligned}$$

Similarly, $AA^\theta = I$.

Hence A is a unitary matrix.

Also

$$A^{-1} = A^\theta = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ -\frac{1}{2}(1+i) & \frac{1}{2}(1+i) \end{bmatrix}$$

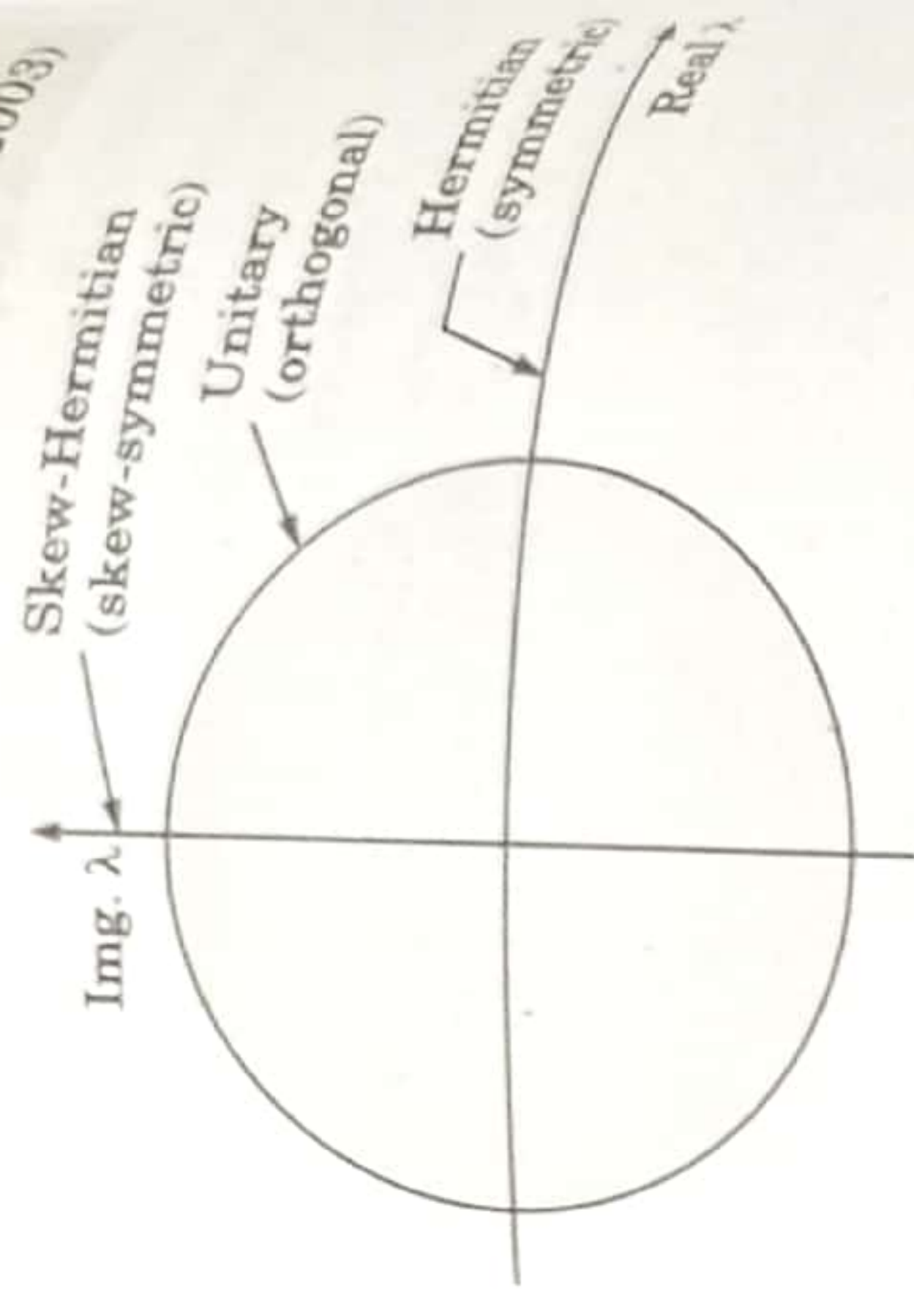


Fig. 2.1. Eigen values of various matrices.

Example 2.55. Given that $A = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$, show that $(I-A)(I+A)^{-1}$ is a unitary matrix.

(Mumbai, 2007)

Solution.

$$I + A = \begin{bmatrix} 1 & 1+2i \\ -1+2i & 1 \end{bmatrix}, |I + A| = 1 - (-1 - 4) = 6$$

$$(I + A)^{-1} = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix} \div 6. \text{ Also } I - A = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$\therefore (I - A)(I + A)^{-1} = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix} \div 6 = \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix}$$

$$\text{Its conjugate-transpose} = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix}$$

$$\therefore \text{Product of (i) and (ii)} = \frac{1}{36} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} \begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 36 & 0 \\ 0 & 36 \end{bmatrix} = I.$$

Hence the result.

PROBLEMS 2.11

1. Prove that every Hermitian matrix can be written as $A + iB$, where A is real and symmetric and B is real and skew-symmetric : (P.T.U., 19)
2. Show that every square matrix can be uniquely expressed as $P + iQ$, where P and Q are Hermitian matrices. (Mumbai, 2008 ; Bhopal, 200)
3. Show that a Hermitian matrix remains Hermitian when transformed by an orthogonal matrix.

4. Show that the matrix $\begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$ is a unitary matrix, if $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1$. (U.P.T.U.,)

5. Show that $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ is a Hermitian matrix.

6. If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, show that A is a Hermitian matrix and iA is a skew-Hermitian matrix.

(Sambalpu)

7. Show that the following matrix is unitary

$$(i) \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \text{ (U.P.T.U., 2002)}$$

$$(ii) \begin{bmatrix} \frac{2+i}{3} & \frac{2i}{3} \\ \frac{2i}{3} & \frac{2-i}{3} \end{bmatrix}$$

(Mum)

Express $A = \begin{bmatrix} 2i & 2+i & 1-i \\ -2+i & -i & 3i \\ -1-i & 3i & 0 \end{bmatrix}$ as $P + iQ$ where P is real and skew-symmetric and Q is real and symmetric. (Mur)

If $S = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}$, where $a = e^{2i\pi/3}$, prove that $S^{-1} = \frac{1}{3}\bar{S}$.

(Kurukshetra, 2006 ; J.N)

Choose the correct answer or fill up the blanks in the following problems.

1. To multiply a matrix by scalar k , multiply

(a) any row by k

(b) every element by k

(c) any column by k

2. If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then A^n is

(a) $\begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$

(b) $\begin{bmatrix} 3^n & (-4)^n \\ 1 & (-1)^n \end{bmatrix}$

(c) $\begin{bmatrix} 1+3n & 1-4n \\ 1+n & 1-n \end{bmatrix}$

(d) $\begin{bmatrix} 1+2n & -4n \\ 1+n & 1-2n \end{bmatrix}$

3. The inverse of the matrix $\begin{bmatrix} -0.5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

(a) $\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -0.25 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

4. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$, then the determinant AB has the value

(a) 4

(b) 8

(c) 16

(d) 32

5. The system of equations $x + 2y + z = 9$, $2x + y + 3z = 7$ can be expressed as

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$

(d) none of the above.

6. If $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} X = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$, then X equals

(a) $\begin{bmatrix} -3 & -14 \\ 4 & 17 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 3 & -14 \\ 4 & -17 \end{bmatrix}$

7. If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then $A(\text{adj } A)$ equals

(a) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 10 & 1 \\ 1 & 10 \end{bmatrix}$

(d) none of the above.

8. If $3x + 2y + z = 0$, $x + 4y + z = 0$, $2x + y + 4z = 0$, be a system of equations, then

(a) it is inconsistent

(b) it has only the trivial solution $x = 0$, $y = 0$, $z = 0$.

(c) it can be reduced to a single equation and so a solution does not exist.

(d) determinant of the matrix of coefficients is zero.

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then

(a) $C = A \cos \theta - B \sin \theta$

(b) $C = A \sin \theta + B \cos \theta$

(c) $C = A \sin \theta - B \cos \theta$

(d) $C = A \cos \theta + B \sin \theta$.

10. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ \beta & \gamma & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$, then

- (a) A is row equivalent to B only when $\alpha = 2$, $\beta = 3$, and $\gamma = 4$
 (b) A is row equivalent to B only when $\alpha \neq 0$, $\beta \neq 0$, and $\gamma = 0$
 (c) A is not row equivalent to B
 (d) A is row equivalent to B for all value of α , β , γ .

11. If $A \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then A is

(a) $\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 1 \\ -1/2 & -1/2 \end{bmatrix}$

12. Matrix has a value. This statement

- (a) is always true
 (c) is false

(b) depends upon the matrices

13. If A is a square matrix such that $AA' = I$, then value of $A'A$ is

(a) A^2

(b) I

(c) A^{-1}

14. If every minor of order r of a matrix A is zero, then rank of A is

(a) greater than r

(b) equal to r

(c) less than or equal to r

(d) less than r .

15. A square matrix A is called orthogonal if

(a) $A = A^2$

(b) $A' = A^{-1}$

(c) $AA^{-1} = I$

16. The rank of matrix $\begin{bmatrix} 2 & -1 & 3 & 1 \\ 1 & 4 & -2 & 1 \\ 5 & 2 & 4 & 3 \end{bmatrix}$ is

17. The sum of the eigen values of a matrix is the of the elements of the principal diagonal.

18. The sum and product of the eigen values of the matrix $\begin{bmatrix} 2 & -3 \\ 4 & -2 \end{bmatrix}$ are and respectively. (Anna, 2009)

19. Inverse of $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ is $\begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & k \\ 2 & 2 & 5 \end{bmatrix}$ then k is

20. Using Cayley-Hamilton theorem, the value of $A^4 - 4A^3 - 5A^2 - A + 2I$ when $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$ is (Anna, 200)

21. If two eigen values of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ are 3 and 15, then the third eigen value is

22. A quadratic form is positive semi-definite when

23. $A_{m \times n}$ and $B_{p \times q}$ are two matrices. When will

(a) $A \cdot B$ exist

(b) $A + B$ exist?

24. The product of the eigen values of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ is

25. The quadratic form corresponding to the diagonal matrix $\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is

(a) $x_1^2 + x_2^2 + \dots + x_n^2$

(b) $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$

(c) $\lambda_1^2 x_1^2 + \lambda_2^2 x_2^2 + \dots + \lambda_n^2 x_n^2$

26. An example of a 3×3 matrix of rank one is

27. The quadratic form corresponding to the symmetric matrix $\begin{bmatrix} 1 & 2 \\ 2 & -4 \end{bmatrix}$ is

28. Solving the equations $x + 2y + 3z = 0$, $3x + 4y + 4z = 0$, $7x + 10y + 12z = 0$, $x = \dots$, $y = \dots$, $z = \dots$

29. The eigen values of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ are

30. A matrix A is idempotent if

31. The rank of the matrix $\begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ is

32. If $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$, then the eigen values of A^2 are

33. The sum of the eigen values of $\begin{bmatrix} 1 & 1 & 3 \\ 3 & -1 & -1 \end{bmatrix}$ is
(iii) 6

(iv) 7

(S.V.T.U.,

(i) -2

34. The maximum value of the rank of a 4×5 matrix is

(Anna,

35. The sum of two eigen values and trace of a 3×3 matrix are equal, then the value of $|A|$ is
36. If the sum of the eigen values of the matrix of the quadratic form is zero, then the nature of the quadratic form is

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37. The eigen values of matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{bmatrix}$ are

38. The eigen values of a triangular matrix are

39. If the product of two eigen values of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is 16, then the third eigen value is

40. If $\lambda_i, i = 1, 2, \dots, n$ are the eigen values of a square matrix A , then the eigen values of A^T are

41. By applying elementary transformations to a matrix, its rank

(a) increases
(b) decreases
(c) does not change

42. If λ is an eigen value of A , then it is an eigen value of B , only if $B = \dots\dots\dots$

43. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$, then eigen values of A^{-1} are

44. The characteristic equation of $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ is

45. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, then eigen values of A^{-1} are

46. Matrix $\begin{bmatrix} x & 2 \\ 1 & x-1 \end{bmatrix}$ is singular for $x = \dots\dots\dots$

47. Every Hermitian matrix can be written as $A + iB$, where A is real and and B is real and

48. The sum and product of the eigen values of $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are and

49. If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then $A^3 = \dots\dots\dots$

50. The product of the eigen values of a matrix is equal to

51. The eigen values of $A = \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$ are the roots of the equation

52. A system of linear non-homogeneous equations is consistent, if and only if the rank of coefficient matrix is equal to rank of
 53. The matrix of the quadratic form $q = 4x^2 - 2y^2 + z^2 - 2xy + 6zx$ is
 54. If $\lambda_1, \lambda_2, \lambda_3$ are the eigen values of a matrix A , then A^3 has the eigen values
 55. If λ is an eigen value of a non-singular matrix A , then the eigen value of A^{-1} is
 56. The matrix corresponding to the quadratic form $x^2 + 2y^2 - 7z^2 - 4xy + 8xz + 5yz$ is
 57. The sum of the squares of the eigen values of $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ is
 58. If the rank of a matrix A is 2, then the rank of A' is
 59. The index and signature of the quadratic form $x_1^2 + 2x_2^2 - 3x_3^2$ are respectively and
 60. The equations $x + 2y = 1, 7x + 14y = 12$ are consistent.
 61. If $\text{rank}(A) = 2, \text{rank}(B) = 3$, then $\text{rank}(AB) = 6$.
 62. Any set of vectors which includes the zero vector is linearly independent.
 63. If λ is an eigen value of a symmetric matrix, then λ is real.
 64. Every square matrix does not satisfy its own characteristic equation.
 65. If λ is an eigen value of an orthogonal matrix, then $1/\lambda$ is also its eigen value.
 66. If the rank of a matrix A is 3, then the rank of $3A^T$ is 1.
 67. The vectors $[1, 1, -1, 1], [1, -1, 2, -1], [3, 1, 0, 1]$ are linearly dependent.
 68. The eigen values of a skew-symmetric matrix are real.
 69. Inverse of a unitary matrix is a unitary matrix.
 70. A is a non-zero column matrix and B is a non-zero row matrix, then rank of AB is one.
 71. The sum of the eigen values of A equals to the trace of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$.